

Asymptotic expansions of inverse functions on a real branch

Power–logarithmic reversion with real exponents, from two Mathematica Stack Exchange questions to a unified theory and a Wolfram Language package

Vladimir Reshetnikov*

September 2026

Abstract

The inverse of $f(x) = x + x^2(1 + \log x)$ on the positive real axis has no Taylor expansion at 0: its expansion is $y - y^2(1 + \log y) + y^3(2 \log^2 y + 5 \log y + 3) - \dots$, with a polynomial in $\log y$ at every power. The inverse of $x + x^{\sqrt{2}}$ has irrational exponents, $y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \dots$. The built-in functions `InverseSeries`, `Series`, `Asymptotic` and `InverseFunction` of the Wolfram Language either fail on such inputs or return the germ at a complex root. The same algebra of finite *power–log expansions* also governs forward asymptotics with irrational exponents, such as $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ at infinity, where `Asymptotic` returns its input unchanged.

We develop the theory completely: the ordered algebra of power–log blocks and its exact truncated arithmetic; a marker form of Lagrange–Bürmann inversion at a positive point with an Euler-operator coefficient formula for every block of $g(y)^r$; selection of the real branch; convergence of the expansion by an implicit-function argument in finitely many small parameters, and an explicit conditional tail majorant by Rouché’s theorem; the remainder after any exponent cutoff as an asymptotic equality with the first omitted block; transport of forward remainders and residuals to the inverse; a theorem showing that finite smoothness of the perturbation suffices; expansions at finite points from either side and at infinity through signed leading powers; powers and logarithms of the inverse; a precision-tracking forward expansion engine; and the boundaries of the theory (leading logarithms and Lambert W , zero gaps, flat terms, oscillations). The Wolfram Language package `AsymptoticInverse` implements all of it with exact real exponents and inert remainder descriptors, and the nine reports that preceded this work are compared, including the first native runs of their test suites.

Contents

1	Introduction	3
1.1	Two questions	3
1.2	What is common to all of this	4
1.3	Summary of results	4
1.4	Conventions	5
2	The power–log scale and its exact arithmetic	6
2.1	Blocks and their order	6
2.2	Exponent semigroups	6

*Written with Claude (Anthropic). The theory synthesises nine independently written reports, listed in Section 14, and adds the extensions needed for the unified package.

2.3	Jets and their arithmetic	7
2.4	Blocks with nonconstant leading coefficient	8
3	Forward expansions	8
3.1	Convergent power–log expansions	8
3.2	The precision-tracking evaluator	10
3.3	The second question	11
4	The model and the real branch	11
4.1	The normal form	11
4.2	Existence and uniqueness of the local inverse	11
4.3	The complex germ returned by <code>InverseFunction</code>	12
5	Lagrange–Bürmann inversion with a marker	12
5.1	The marker theorem	13
5.2	Consequences	13
6	The coefficient theorem	14
6.1	Statement	14
6.2	First orders and cross terms	15
6.3	Degrees and leading logarithms	15
6.4	Constant coefficients: Fuss–Catalan numbers	16
6.5	Formal existence, uniqueness, and the two iterative engines	17
7	Convergence	17
7.1	Convergence in finitely many small parameters	18
7.2	An explicit conditional majorant	18
8	Remainders after a cutoff	19
8.1	Exponent cutoff: the first omitted block	19
8.2	Depth truncation	21
8.3	Numerical illustration	21
9	Transport of remainders and finite smoothness	21
9.1	From the forward function to the inverse	22
9.2	From a residual to the error of the inverse	23
9.3	Finite smoothness suffices	23
10	Finite endpoints, infinity, and powers of the inverse	25
10.1	Local variables	25
10.2	The theorem at an arbitrary endpoint	25
10.3	Powers of the inverse and the original variable	26
10.4	Examples	26

11 Worked examples	27
11.1 The two examples of the first question, once more	27
11.2 Nonunit leading power	27
11.3 Several gaps	27
11.4 Automatically expanded forward functions	28
11.5 Symbolic data	28
11.6 Powers and logarithms of the inverse	28
12 Boundaries of the theory	28
12.1 A leading logarithm: Lambert W and the log log scale	28
12.2 Zero gaps: corrections of relative size $1/\log$	29
12.3 Flat terms	29
12.4 Oscillatory coefficients	29
12.5 Symbolic exponents and nonuniformity	30
12.6 Nested powers of the logarithm	30
13 The package	30
13.1 Data structures and exact comparisons	30
13.2 The engines	31
13.3 Checks	31
13.4 Cost	32
14 The nine reports	32
14.1 Agreement	32
14.2 Individual contributions absorbed here	32
14.3 Errors found	33
14.4 Native execution	33

1 Introduction

1.1 Two questions

The first question [1] asks for an asymptotic expansion, as $y \rightarrow 0^+$, of the inverse of

$$f_1(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1.1)$$

on the branch that maps $(0, \infty)$ to itself. The natural attempts fail in instructive ways. `AsymptoticSolve[x + x^2 (1 + Log[x]) == y, {x}, y -> 0]` returns unevaluated; `InverseSeries` returns unevaluated because the forward series has a coefficient containing $\log x$; and

```
Series[InverseFunction[Function[x, x + x^2 (1 + Log[x])]][y], {y, 0, 2}]
```

returns the perfectly correct Taylor germ

$$c + \frac{y}{c-1} - \frac{3 + 2W_{-1}(-e)}{2(c-1)^3} y^2 + \mathcal{O}(y^3), \quad c = e^{W_{-1}(-e)-1} \approx -0.1178 - 0.5332i, \quad (1.2)$$

of the inverse at a *different* preimage of 0: the nonzero complex root c of $1 + x(1 + \log x) = 0$, where f_1 is analytic and $f_1'(c) = c - 1 \neq 0$ (Section 4). The real branch has no Taylor expansion at all. Its expansion is

$$g_1(y) = y - y^2(1 + L) + y^3(2L^2 + 5L + 3) - y^4\left(5L^3 + \frac{41}{2}L^2 + 27L + \frac{23}{2}\right) + y^5\left(14L^4 + \frac{241}{3}L^3 + \frac{335}{2}L^2 + 151L + \frac{299}{6}\right) + \mathcal{O}(y^6(1 + |L|)^5), \quad (1.3)$$

with $L = \log y$: every power of y carries a polynomial in $\log y$, of degree growing with the power, so the scale of the expansion is neither a Taylor nor a Puiseux scale. The question also asks, as a second example, for

$$g_2(y) = y - y^{\sqrt{2}} + \sqrt{2}y^{2\sqrt{2}-1} - \frac{6 - \sqrt{2}}{2}y^{3\sqrt{2}-2} + \left(\frac{17\sqrt{2}}{3} - 4\right)y^{4\sqrt{2}-3} + \mathcal{O}(y^{5\sqrt{2}-4}), \quad (1.4)$$

the inverse of $f_2(x) = x + x^{\sqrt{2}}$: no logarithms this time, but exponents in the semigroup $1 + (\sqrt{2} - 1)\mathbb{N}$, which no rational grid contains.

The second question [2] is about forward asymptotics. `Asymptotic[(1 + x + x^Sqrt[2])^Sqrt[2], x -> Infinity]` correctly returns x^2 , but with `SeriesTermGoal -> 7` it returns the input unchanged. The expansion is

$$(1 + x + x^{\sqrt{2}})^{\sqrt{2}} = x^2 + \sqrt{2}x^{3-\sqrt{2}} + \sqrt{2}x^{2-\sqrt{2}} + \left(1 - \frac{1}{\sqrt{2}}\right)x^{4-2\sqrt{2}} + (2 - \sqrt{2})x^{3-2\sqrt{2}} + \left(\frac{2\sqrt{2}}{3} - 1\right)x^{5-3\sqrt{2}} + \frac{13-9\sqrt{2}}{12}x^{6-4\sqrt{2}} + \mathcal{O}(x^{7-5\sqrt{2}}), \quad (1.5)$$

and the inverse of this function, as $y \rightarrow \infty$, begins $\sqrt{y} - \frac{1}{\sqrt{2}}y^{(2-\sqrt{2})/2} - \frac{1}{\sqrt{2}}y^{(1-\sqrt{2})/2} + \dots$.

1.2 What is common to all of this

All these expansions are finite or infinite sums of *power-log blocks* $w^\beta P(\log w)$, where $w \rightarrow 0^+$ is a local variable ($w = y$, $w = 1/x$, $\dots$), β is a real exponent, and P is a polynomial. The exponents lie in a finitely generated additive semigroup, which is locally finite even when it is dense in $\mathbb{R}$, so every truncation is a finite computation; the logarithmic polynomials are handled by the Euler identity $\frac{d}{dw}[w^\beta P(\log w)] = w^{\beta-1}(\beta + \frac{d}{dL})P(L)$; and remainders must keep their logarithmic factor, because $y^3(2\log^2 y + 5\log y + 3)$ is not $\mathcal{O}(y^3)$.

This article gives the complete theory of this algebra as far as it is needed to compute and to justify such expansions, forward and inverse, with proofs, and describes the package `AsymptoticInverse` that implements it. It grew out of nine independently written reports on the first question (Section 14 and [12]), which agree on the core of the theory and each contribute something the others lack; here their contents are unified, the proofs are completed, and the theory is extended to finite endpoints from either side, to infinity, to powers and logarithms of the inverse, and to the forward expansion problem of the second question.

1.3 Summary of results

- **Scale** (Section 2). Power-log blocks are totally ordered by size, the finitely generated exponent semigroups are locally finite, and the truncated arithmetic (products, real powers, logarithms, exponentials, composition with polynomials in the logarithm) is exact and terminates. The single primitive is the composition recurrence $Q_{k+1} = ((a - k)Q_k + Q'_k)/(k + 1)$ for $(1 + t)^a P(L + \log(1 + t))$.

- **Forward expansions** (Section 3). Expressions built from constants, the variable, arithmetic, real powers, logarithms, bounded exponentials and analytic functions of expressions with a limit have convergent power–log expansions; a precision-tracking evaluator produces them with a rigorous remainder class, which answers the second question.
- **Branch** (Section 4). For $f(x) = ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$ with $a, p \neq 0$, $\delta_i > 0$, the real branch is the one with $x > 0$ and $x/(y/a)^{1/p} \rightarrow 1$; it is unique, and the complex germ (1.2) is explained.
- **Lagrange–Bürmann with a marker** (Section 5). A residue proof of $\Phi(u) = \Phi(q) + \sum_N \frac{(-\eta)^N}{N!} \partial_q^{N-1} [\Phi'(q)h(q)^N]$ at a positive point, valid analytically and formally, with the transport of observables and the general-core form.
- **Coefficients** (Section 6). For every multi-index $\mathbf{k}$ of weight σ , the block of $g(y)^r$ at $z^{r+\sigma}$, $z = (y/a)^{1/p}$, is $\frac{(-1)^{|\mathbf{k}|} r}{p^{|\mathbf{k}|} \mathbf{k}!} \prod_{j=1}^{|\mathbf{k}|-1} (\mathcal{D} + r + \sigma + pj) \prod_i B_i(L)^{k_i}$, $\mathcal{D} = d/dL$; equal weights are added. Degrees, leading logarithms (Catalan numbers), closed forms for constant coefficients (Fuss–Catalan), $\log g$, and the formal uniqueness through the triangular equation, Picard iteration and Newton iteration.
- **Convergence** (Section 7). The expansion converges absolutely for small positive y after substituting finitely many small parameters $z^{\delta_i}(\log z)^j$, with coefficient bounds $|C_{\mathbf{k}}(L)| \leq MK^{|\mathbf{k}|}(1 + |L|)^{D(\mathbf{k})}$; an explicit conditional tail majorant follows from Rouché’s theorem.
- **Remainders** (Section 8). Truncation below any exponent cutoff leaves $z^{1+\sigma_*} C_{\sigma_*}(L) + o(z^{1+\sigma_*})$ where σ_* is the first omitted weight and C_{σ_*} the complete coefficient there, computed from the one-step boundary of the retained multi-index set; depth truncation has its own bound.
- **Transport** (Section 9). Forward remainders and residuals are transported to the inverse with the correct scaling z^{1-p} ; a residual bracket gives existence and a two-sided error bound; and a theorem with a complete proof shows that finite smoothness of the perturbation, with derivative bounds in the power–log scale, suffices for the marker formula with an explicit remainder, which covers perturbations such as $x^2 \sin(\log x)$.
- **Endpoints** (Section 10). Finite points from either side and $\pm\infty$ reduce to the same engine through a local variable and a signed leading power $p \neq 0$; the limit y_0 may be finite or infinite; powers $(x - x_0)^r$ and x^r of the inverse are obtained directly. Worked examples include the Wright omega function and the inverse of (1.5).
- **Boundaries** (Section 12). Leading logarithms force a Lambert- W coordinate and a log log scale; zero gaps force $1/\log y$; flat and oscillatory terms; symbolic exponents and depth truncation; what an extension needs.
- **Package and reports** (Sections 13, 14). Data structures, exact comparison of algebraic exponents, the two engines, precision rules, checks; and the comparison of the nine reports, whose Wolfram packages were run natively for the first time.

1.4 Conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$. For a local variable $w \rightarrow 0^+$ we write $L = \log w$ and $M(w) = 1 + |\log w|$. $\mathcal{O}$ and o refer to $w \rightarrow 0^+$ with all other data fixed; no uniformity in parameters is asserted unless stated. $\mathcal{D} = d/dL$ acts on polynomials in L . For a multi-index $\mathbf{k} = (k_1, \dots, k_m) \in \mathbb{N}^m$ we write $|\mathbf{k}| = \sum k_i$, $\mathbf{k}! = \prod k_i!$, and $\mathbf{k} \cdot \boldsymbol{\delta} = \sum k_i \delta_i$. Real powers of positive quantities are $w^\beta = e^{\beta \log w}$; complex powers are never used on the real branch. Cutoffs are *exclusive*: a cutoff q retains the blocks of exponent $< q$. Wolfram Language names are set in typewriter font.

2 The power–log scale and its exact arithmetic

2.1 Blocks and their order

Throughout, w denotes a positive real variable tending to 0^+ , $L = \log w \rightarrow -\infty$ and $M(w) = 1 + |\log w|$.

Definition 2.1 (Blocks). A *power–log block* is an expression $w^\beta P(L)$ with $\beta \in \mathbb{R}$ and $P \in \mathbb{R}[L]$, $P \neq 0$. Its *exponent* is β and its *logarithmic degree* is $\deg P$.

Lemma 2.2 (Power beats logarithm). *For every $\varepsilon > 0$ and every real b , $w^\varepsilon M(w)^b \rightarrow 0$ as $w \rightarrow 0^+$. Consequently, if $\beta < \gamma$ and $P, Q \neq 0$, then $w^\gamma Q(L) = o(w^\beta P(L))$; and within a fixed exponent, a higher logarithmic degree dominates: $w^\beta L^j = o(w^\beta L^k)$ for $j < k$.*

Proof. Put $w = e^{-t}$, $t \rightarrow +\infty$: $w^\varepsilon M(w)^b = e^{-\varepsilon t} (1+t)^b \rightarrow 0$. For the second claim the ratio is $w^{\gamma-\beta} Q(L)/P(L)$, and $|Q(L)/P(L)| \leq CM(w)^{\deg Q}$ for w small because $P(L)$ has no zeros for L sufficiently negative and $|P(L)| \geq c|L|^{\deg P} \geq c$ there; the first claim finishes. The third claim is clear. $\square$

Blocks are therefore totally ordered by size: by increasing exponent, and at equal exponents by decreasing logarithmic degree. A finite sum of blocks with distinct exponents is asymptotic to its block of least exponent (whole polynomial included), and vanishes identically only if every block does. This is why truncations are taken by *complete blocks*: cutting a block inside its polynomial would drop terms larger than the retained terms of the next exponent.

Definition 2.3 (Remainder classes). For $\beta \in \mathbb{R}$ and an integer $k \geq 0$ we write $R(w) = \mathcal{O}_{\beta,k}$, or in the package `PowerLogRemainder[w,  $\beta$ , k]`, for $R(w) = \mathcal{O}(w^\beta M(w)^k)$ as $w \rightarrow 0^+$.

By Lemma 2.2, $\mathcal{O}_{\beta,k} \subseteq \mathcal{O}_{\beta',k'}$ whenever $\beta > \beta'$, or $\beta = \beta'$ and $k \leq k'$; the class with the smaller exponent, and at equal exponents the larger degree, is the coarser one. A block $w^\beta P(L)$ with $\deg P = k$ belongs to $\mathcal{O}_{\beta,k}$ and to no $\mathcal{O}_{\beta,k'}$ with $k' < k$. Sums and products of remainder classes obey

$$\mathcal{O}_{\beta,k} + \mathcal{O}_{\beta',k'} = \mathcal{O}_{\min(\beta,\beta'),k''}, \quad \mathcal{O}_{\beta,k} \cdot \mathcal{O}_{\beta',k'} = \mathcal{O}_{\beta+\beta',k+k'}, \quad (2.1)$$

where k'' is the degree attached to the smaller exponent, and the larger of the two when the exponents coincide.

2.2 Exponent semigroups

Lemma 2.4 (Local finiteness). *Let $\delta_1, \dots, \delta_m > 0$ and $S = \{\mathbf{k} \cdot \boldsymbol{\delta} : \mathbf{k} \in \mathbb{N}^m\}$. For every real H the set $\{\mathbf{k} \in \mathbb{N}^m : \mathbf{k} \cdot \boldsymbol{\delta} \leq H\}$ is finite, hence $S \cap (-\infty, H]$ is finite, and every $\sigma \in S$ has only finitely many representations $\sigma = \mathbf{k} \cdot \boldsymbol{\delta}$.*

Proof. If $\mathbf{k} \cdot \boldsymbol{\delta} \leq H$ then $k_i \leq H/\delta_i$ for each i . $\square$

When the gaps are incommensurable the *group* generated by them is dense in $\mathbb{R}$ (for instance $\mathbb{Z} + \sqrt{2}\mathbb{Z}$), but the semigroup of nonnegative combinations has only finitely many elements below any bound. This is the whole reason why expansions with irrational exponents can be computed term by term: the support of every expansion in this article lies in a translate $\beta_0 + S$ of such a semigroup. Following [11], a subset of $\mathbb{R}$ that is finite below every bound is called *left-finite*; the semigroups S are left-finite, and so are their translates and finite unions.

2.3 Jets and their arithmetic

Definition 2.5 (Jets). Fix gaps as above and $\beta_0 \in \mathbb{R}$. A *jet* (of cutoff H) is a finite sum $U = \sum_{\sigma \in T} w^\sigma C_\sigma(L)$ with $T \subset \beta_0 + S$ finite, $C_\sigma \in \mathbb{R}[L]$, all $\sigma < H$; equal exponents are merged and zero polynomials are discarded. Its *valuation* $\text{val } U$ is its least exponent ($+\infty$ for the empty jet). The jet algebra $\mathcal{P}_{<H}$ consists of jets truncated below H , with the sum and the product computed exactly and then truncated.

The jet algebra is the quotient of the ring of finite sums of blocks by the ideal of blocks of exponent $\geq H$. In the package a jet is a sorted list of pairs $\{\sigma, C_\sigma\}$ with σ an exact real number in canonical form (`RootReduce` for algebraic numbers) and C_σ a polynomial in a private symbol ℓ ; two exponents are equal if and only if their canonical forms coincide, and they are ordered exactly (Section 13).

Differentiation

Lemma 2.6 (Euler identity). For $\beta \in \mathbb{R}$, $P \in \mathbb{R}[L]$ and $r \in \mathbb{N}$,

$$\frac{d}{dw}[w^\beta P(L)] = w^{\beta-1}(\beta + \mathcal{D})P(L), \quad \frac{d^r}{dw^r}[w^\beta P(L)] = w^{\beta-r} \prod_{j=0}^{r-1} (\beta - j + \mathcal{D})P(L). \quad (2.2)$$

The operators $\beta - j + \mathcal{D}$ commute, and $(\beta + \mathcal{D})$ preserves $\deg P$ when $\beta \neq 0$ and lowers it by one when $\beta = 0$.

Proof. $\frac{d}{dw}[w^\beta P(\log w)] = \beta w^{\beta-1}P + w^\beta P'(\log w)/w$. Iterate. $\square$

The identity says that differentiation acts on a block as the scalar β plus $\mathcal{D}$ on the polynomial; the exponent bookkeeping and the polynomial arithmetic separate completely. In the coefficient formulas below the operators $\mathcal{D} + c$ with $c > 0$ appear, and their action on the coefficient vector $(z_0, \dots, z_d)$ of $P = \sum z_s L^s$ is bidiagonal: $(c + \mathcal{D})P = \sum_s (cz_s + (s+1)z_{s+1})L^s$.

Unit series

Let U be a jet with $\text{val } U > 0$. Then U^k has valuation $\geq k \text{val } U$, so in $\mathcal{P}_{<H}$ only finitely many powers of U are nonzero, and every power series $\Phi(t) = \sum_k \phi_k t^k$ can be substituted:

$$\Phi(U) = \sum_{k < H/\text{val } U} \phi_k U^k \in \mathcal{P}_{<H}. \quad (2.3)$$

In particular $(1 + U)^a = \sum_k \binom{a}{k} U^k$ for every real a (with $\binom{a}{k} = a(a-1)\cdots(a-k+1)/k!$), $\log(1 + U) = \sum_{k \geq 1} (-1)^{k+1} U^k / k$, $e^U = \sum_k U^k / k!$, and $\phi(c + U) = \sum_k \phi^{(k)}(c) U^k / k!$ for any ϕ analytic at c . These are identities between finite sums of blocks modulo exponents $\geq H$; their analytic content is discussed in Section 7.

Composition with the logarithm

The one operation that mixes the exponent bookkeeping with the polynomial coefficients is the substitution of a jet into the logarithm: if w is replaced by $w(1 + U)$ then L becomes $L + \log(1 + U)$, and a block $w^a P(L)$ becomes $w^a (1 + U)^a P(L + \log(1 + U))$. The following recurrence computes this without expanding $\log(1 + U)$ separately.

Lemma 2.7 (Composition recurrence). *Let $P \in \mathbb{R}[L]$ and $a \in \mathbb{R}$. For $|t| < 1$,*

$$(1+t)^a P(L + \log(1+t)) = \sum_{k \geq 0} Q_k(L) t^k, \quad Q_0 = P, \quad Q_{k+1} = \frac{(a-k)Q_k + Q'_k}{k+1}, \quad (2.4)$$

with $Q_k \in \mathbb{R}[L]$, $\deg Q_k \leq \deg P$, and, when $a \in \mathbb{N}$ and P is constant, $Q_k = 0$ for $k > a$. Consequently, for a jet U of positive valuation, $(1+U)^a P(L + \log(1+U)) = \sum_k Q_k(L) U^k$ in $\mathcal{P}_{<H}$.

Proof. Put $F(t) = (1+t)^a P(L + \log(1+t))$; it is analytic in $|t| < 1$ with values in $\mathbb{R}[L]$, so $F = \sum Q_k t^k$ with $Q_k \in \mathbb{R}[L]$ and $Q_0 = P$. Differentiating, $(1+t)\partial_t F = aF + \partial_L F$, because ∂_t acting on the two factors gives $a(1+t)^{a-1}P + (1+t)^{a-1}P'$. Comparing coefficients of t^k in $(1+t)\sum (k+1)Q_{k+1}t^k = \sum (aQ_k + Q'_k)t^k$ gives $(k+1)Q_{k+1} + kQ_k = aQ_k + Q'_k$. The degree does not increase, and for $a \in \mathbb{N}$ and constant P the factor $(a-k)$ annihilates Q_{a+1} and hence all later Q_k (for nonconstant P the derivative keeps the sequence alive: $\log(1+t)$ is not a polynomial). The jet statement follows because substitution of U for t is a ring homomorphism on finite sums. $\square$

Remark 2.8. The recurrence is the whole composition calculus needed here: the forward model $ax^p(1 + \sum x^{\delta_i} B_i(\log x))$ evaluated at $x = z(1+U)$ is $az^p(1+U)^p(1 + \sum_i z^{\delta_i}(1+U)^{\delta_i} B_i(L + \log(1+U)))$, each factor of which is an instance of (2.4) (with $P = 1$ for the first). Its derivative is obtained from the same lemma with a replaced by $a-1$ and P by $aP + P'$, since $\frac{d}{dx}[x^a P(\log x)] = x^{a-1}(aP + P')(\log x)$.

2.4 Blocks with nonconstant leading coefficient

Everything above allows the polynomial in the leading block to be nonconstant. Two operations do not: taking a real non-integer power or a logarithm of a sum whose leading block is $w^\beta P(L)$ with $\deg P \geq 1$ produces $P(L)^a$ or $\log P(L)$, which are not polynomials in L . Such expressions live in larger scales (rational functions of L , or iterated logarithms, Section 12) and are outside the class treated here; the package reports them as unsupported. Nonnegative integer powers of any jet are computed by repeated multiplication and stay in the class.

3 Forward expansions

Before inverting anything we need the forward function in the normal form $ax^p(1 + \sum_i x^{\delta_i} B_i(\log x))$, and the second question [2] is a forward problem in its own right. This section shows that a large class of expressions has convergent power-log expansions, and describes the precision-tracking evaluator that computes them with a rigorous remainder class.

3.1 Convergent power-log expansions

Definition 3.1. A function F on $(0, w_0)$ has a *convergent power-log expansion* with support in $\beta_0 + S$ (S a finitely generated positive semigroup with gaps $\delta_1, \dots, \delta_m$) if there are polynomials $C_\sigma \in \mathbb{R}[L]$ and constants $M, K, d \geq 0$ such that

$$F(w) = \sum_{\sigma \in S} w^{\beta_0 + \sigma} C_\sigma(L) \quad \text{for } 0 < w < w_1, \quad |C_\sigma(L)| \leq M K^{n(\sigma)} M(w)^{dn(\sigma)} \quad (L \leq \log w_1), \quad (3.1)$$

where $n(\sigma)$ is the least $|\mathbf{k}|$ with $\mathbf{k} \cdot \boldsymbol{\delta} = \sigma$ and the series converges absolutely. We write $F \in \mathcal{E}$.

The coefficient bound makes the tail after any cutoff controllable (Lemma 3.2 below), and it is stable under all the operations we need.

Lemma 3.2 (Tail of a convergent expansion). *Let $F \in \mathcal{E}$ as in (3.1), $\delta_* = \min_i \delta_i$, and $H > 0$. Then*

$$F(w) - \sum_{\sigma < H} w^{\beta_0 + \sigma} C_\sigma(L) = \mathcal{O}(w^{\beta_0 + H} M(w)^{Nd}), \quad N = \lceil H/\delta_* \rceil. \quad (3.2)$$

More precisely, the omitted part equals $w^{\beta_0 + \sigma_} C_{\sigma_*}(L) + o(w^{\beta_0 + \sigma_*})$ where $\sigma_* = \min\{\sigma \in S : \sigma \geq H\}$.*

Proof. Split the omitted terms into those with $n(\sigma) \leq N - 1$ and the rest. The first group is finite (Lemma 2.4) and each of its terms is $\mathcal{O}(w^{\beta_0 + H} M^{(N-1)d})$ since $\sigma \geq H$ and $\deg C_\sigma \leq dn(\sigma)$ follows from the bound. For the rest, the number of multi-indices with $|\mathbf{k}| = n$ is at most m^n , and $\sigma \geq n\delta_*$, so the sum is bounded by $Mw^{\beta_0} \sum_{n \geq N} (mKw^{\delta_*} M^d)^n = \mathcal{O}(w^{\beta_0 + N\delta_*} M^{Nd})$, and $N\delta_* \geq H$. The refined statement follows by applying this with H replaced by any $H' \in (\sigma_*, \sigma_* + \delta_*)$, which shows that everything beyond σ_* is $\mathcal{O}(w^{\beta_0 + H'} M^{N'd}) = o(w^{\beta_0 + \sigma_*})$ by Lemma 2.2. $\square$

Theorem 3.3 (Closure). *The class $\mathcal{E}$ contains the constants and w and is closed under sums, products, and the following operations, with the supports and gaps indicated.*

1. *If $F \in \mathcal{E}$ has leading block cw^β with a constant $c > 0$, then $F^a \in \mathcal{E}$ for every real a and $\log F \in \mathcal{E}$; the support of F^a is $a\beta + S'$ and that of $\log F$ is $\{0\} \cup S'$ (with the block $\log c + \beta L$ at exponent 0), where S' is generated by the gaps of $F/(cw^\beta) - 1$.*
2. *If $F \in \mathcal{E}$ is bounded, i.e. $\beta_0 \geq 0$, and its exponent-0 block is $c + kL$ with $k \in \mathbb{R}$, then $e^F \in \mathcal{E}$ with leading block $e^c w^k$.*
3. *If $F \in \mathcal{E}$ has $\beta_0 \geq 0$ and constant exponent-0 block c , and ϕ is analytic at c , then $\phi \circ F \in \mathcal{E}$.*

Proof. Sums and products: the union and the sum of two finitely generated semigroups are finitely generated, absolutely convergent series may be multiplied, and the coefficient bound of a product follows from $M^{dn} M^{d'n'} \leq M^{\max(d,d')(n+n')}$ and the finiteness of the number of representations. For (1)–(3) write the relevant part of F as $cw^\beta(1 + U)$ or $c + U$ with $U = \sum_{\sigma > 0} w^\sigma C_\sigma(L)$ and $\text{val } U > 0$. Group the terms of U by multi-index and logarithmic degree and introduce the finitely many *small parameters* $T_{\mathbf{k},j}$ standing for $w^{\mathbf{k} \cdot \delta} L^j$; then $U = \sum_{\mathbf{k},j} T_{\mathbf{k},j} c_{\mathbf{k},j}$ is, for the purposes of the operation, a convergent power series in the T 's (this is where the coefficient bound is used: the series in the T 's converges in a polydisc $|T_{\mathbf{k},j}| < \rho$ because $\sum MK^n \rho^n (d+1)^n < \infty$ for small ρ). The map $T \mapsto (1 + U)^a$, $\log(1 + U)$, e^U , or $\phi(c + U)$ is analytic near $T = 0$, hence has a convergent multivariate Taylor series whose coefficients obey Cauchy estimates $|c_\nu| \leq M' \rho'^{-|\nu|}$. Substituting $T_{\mathbf{k},j} = w^{\mathbf{k} \cdot \delta} L^j$ and regrouping the absolutely convergent double series by total weight gives (3.1) with the stated support; the number of monomials ν contributing to a given multi-index is at most $(d+1)^{|\nu|}$, which is absorbed into K , and each such monomial has logarithmic degree at most $d|\nu|$. $\square$

The theorem is the analytic backbone of every forward and inverse expansion in this article: an expression built by finitely many of these operations has a convergent expansion, and every truncation is a genuine asymptotic expansion by Lemma 3.2. The hypotheses of (1)–(3) are also necessary within the power–log class: a non-integer power or a logarithm of a leading block $w^\beta P(L)$ with $\deg P \geq 1$, an exponential of an unbounded expression (which produces $e^{-1/w}$ -type terms), or an analytic function applied to $c + kL + \dots$ with $k \neq 0$ (which produces $\sin(\log w)$ -type oscillations) leave the class (Section 12).

3.2 The precision-tracking evaluator

The package computes forward expansions by structural recursion over the expression tree, carrying for every subexpression a triple (T, P, D) : a jet T and the assertion

$$\text{subexpression} = T + \mathcal{O}(w^P M(w)^D), \quad P \in \mathbb{R} \cup \{+\infty\}, \quad (3.3)$$

where $P = +\infty$ means that T is exact. A global working order K_w bounds the number of terms of every unit series that is expanded. The rules, all justified by (2.1) and Lemma 3.2, are:

- **constants** and w are exact; **Plus**: $T_1 + T_2$ truncated below $\min(P_1, P_2)$, with the degree of the smaller precision (the maximum when the precisions coincide);
- **Times**: $T_1 T_2$ truncated below $\min(P_1 + \text{val } T_2, P_2 + \text{val } T_1)$, with degrees $D_1 + \text{deg}_{\text{lead}} T_2$ and $D_2 + \text{deg}_{\text{lead}} T_1$ (an empty jet contributes its own P and D);
- **Power** with an exact real exponent a : nonnegative integers by repeated multiplication (exact); otherwise the leading block cw^β of T must have constant $c > 0$ (or $c < 0$ with integer a); with $U = T/(cw^\beta) - 1$, relative precision $P - \beta$ and relative cutoff $c_{\text{rel}} = \min(P - \beta, K_w - a\beta)$, the result is $c^a w^{a\beta} \sum_k \binom{a}{k} U^k$ with precision $a\beta + c_{\text{rel}}$ and degree $\lceil c_{\text{rel}} / \text{val } U \rceil \cdot \text{deg}_{\text{max}} U$ from the truncation of the binomial series (Lemma 3.2 applied to the finite jet U), combined with D by (2.1);
- **Log**: the leading block must be cw^β with constant $c > 0$; the result is $\log c + \beta L + \log(1 + U)$;
- **Exp**: T must have no negative exponents and its exponent-0 block must be $c + kL$ with exact numeric k ; the result is $e^{cw^k} \exp(U)$;
- an analytic head ϕ (**Sin**, **ArcTan**, **BesselJ**[0, .], ...) applied to $T = c + U$ with constant c : $\sum_k \phi^{(k)}(c) U^k / k!$ with the Taylor coefficients obtained from **Series**[$\phi[c + \mathbf{t}]$, $\{\mathbf{t}, 0, \mathbf{N}\}$];
- any other subexpression is expanded by **Series**[**expr**, $\{\mathbf{w}, 0, \mathbf{n}\}$, **Assumptions** $\rightarrow \mathbf{w} > 0$]; the coefficients (which may contain $\log w$) and the irrational-power monomials that **Series** leaves outside its **SeriesData** are parsed as jets, the precision is the order of the 0 term, and the logarithmic degree D is read from the first omitted block obtained by expanding one order further.

The whole expression is re-expanded with a larger K_w until the final precision reaches the requested cutoff. The result is reported as the retained blocks plus $\mathcal{O}_{\sigma_*, k_*}$, where σ_* is the exponent of the first omitted block if one lies below the final precision (its complete coefficient having been computed), and otherwise the precision pair (P, D) itself.

Proposition 3.4 (Soundness). *If every leaf and every **Series** fallback satisfies its asserted (3.3), then so does the root, and the reported remainder class is valid.*

Proof. Induction over the tree with (2.1) for sums and products, Lemma 3.2 for the truncated unit series (the finite jet U generates a finitely generated semigroup with $\delta_* = \text{val } U$ and $d = \text{deg}_{\text{max}} U$, and Theorem 3.3 makes the unit series a convergent expansion), and the mean value estimate $\Phi(1 + U) - \Phi(1 + U_{\text{known}}) = \mathcal{O}(U - U_{\text{known}})$ for the contribution of the unknown part of the argument. The final remainder statement is Lemma 3.2's refined form. $\square$

The only hypothesis that the package cannot verify is the last bullet: when **Series** is used, the assertion that the omitted part is $\mathcal{O}(w^\rho M^k)$ with k the degree of the first omitted block is the assumption that **Series** produces a genuine asymptotic expansion in the power-log scale, which is the case for the functions it handles (analytic functions of w , w^α and $\log w$). Exact finite expressions never use the fallback and their remainders are proved.

3.3 The second question

For $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ at $x \rightarrow +\infty$ the local variable is $w = 1/x$ and

$$F = w^{-\sqrt{2}}(1 + w^{\sqrt{2}-1} + w^{\sqrt{2}})^{\sqrt{2}} = w^{-2} \sum_{k \geq 0} \binom{\sqrt{2}}{k} (w^{\sqrt{2}-1} + w^{\sqrt{2}})^k,$$

an instance of item (1) of Theorem 3.3 with gaps $\sqrt{2} - 1$ and $\sqrt{2}$. The exponents of x are $2 - (\sqrt{2} - 1)k_1 - \sqrt{2}k_2$; ordering the semigroup elements gives (1.5), and the first omitted exponent is $7 - 5\sqrt{2}$ (multi-index $(5, 0)$). The workaround in the question, which peels off one leading term per limit computation, produces the same seven terms; the evaluator produces them in a fraction of a second together with the remainder $\mathcal{O}(x^{7-5\sqrt{2}})$, and any number of further terms.

4 The model and the real branch

4.1 The normal form

Let $u \rightarrow 0^+$ be the local variable (later $u = x - x_0$, $x_0 - x$, $1/x$ or $-1/x$). The central object is a finite *power-log germ*

$$f(u) = y_0 + a u^p \left(1 + \sum_{i=1}^m u^{\delta_i} B_i(\log u) \right), \quad a \neq 0, \quad p \neq 0, \quad \delta_i > 0, \quad B_i \in \mathbb{R}[L] \setminus \{0\}, \quad (4.1)$$

with $y_0 \in \mathbb{R}$ when $p > 0$ and $y_0 = 0$ when $p < 0$ (a constant term is then a correction with gap $-p$). Equal gaps are merged. The *leading block* au^p has a constant coefficient; a logarithmic factor in the leading block ($u \log u$, $u(\log u)^{-1}$) changes the problem (Section 12). Everything below is stated for this exact finite germ; functions that are only known through a truncated expansion are handled in Section 9.

The *uniformizer* is

$$z = \left(\frac{v}{a} \right)^{1/p}, \quad v = y - y_0 \quad (p > 0), \quad v = y \quad (p < 0), \quad L = \log z = \frac{\log(v/a)}{p}, \quad (4.2)$$

so that $z \rightarrow 0^+$ in every case: for $p > 0$ because $v \rightarrow 0$ with $v/a > 0$, for $p < 0$ because $v \rightarrow \pm\infty$ with the sign of a . The role of z is that the inverse satisfies $u \sim z$.

4.2 Existence and uniqueness of the local inverse

Theorem 4.1 (Positive local branch). *For the germ (4.1) there is $\varepsilon > 0$ such that f is strictly monotone on $(0, \varepsilon)$, with*

$$f(u) - y_0 \sim a u^p, \quad f'(u) \sim a p u^{p-1} \quad (u \rightarrow 0^+). \quad (4.3)$$

Its inverse g , defined on the image of $(0, \varepsilon)$, is the unique function with $g > 0$, $g \rightarrow 0$ and $f(g(y)) = y$ there, and it satisfies

$$g(y) \sim z, \quad g'(y) \sim \frac{z}{p v} \quad (z \rightarrow 0^+). \quad (4.4)$$

Proof. Write $R(u) = \sum_i u^{\delta_i} B_i(\log u)$. By Lemma 2.2, $R(u) \rightarrow 0$ and, since $uR'(u) = \sum_i u^{\delta_i} (\delta_i B_i + B'_i)(\log u)$ by Lemma 2.6, also $uR'(u) \rightarrow 0$. Hence

$$f'(u) = a p u^{p-1} \left(1 + R(u) + \frac{uR'(u)}{p} \right) = a p u^{p-1} (1 + o(1)),$$

which has the sign of ap on some $(0, \varepsilon)$; this gives monotonicity and (4.3). The inverse exists on the image, tends to 0 because f is a homeomorphism of $(0, \varepsilon)$ onto its image, and is the only solution of $f(u) = y$ in $(0, \varepsilon)$. Substituting $u = g(y)$ into $f(u) - y_0 \sim au^p$ gives $v \sim ag(y)^p$, i.e. $g \sim z$; and $g' = 1/f'(g) \sim 1/(apg^{p-1}) = g/(pag^p) \sim z/(pv)$. $\square$

The theorem is local. Whether the local branch extends to a global inverse is a separate question, answered for the two examples of [1] as follows.

Proposition 4.2 (Global invertibility of the examples). $f_1(x) = x + x^2(1 + \log x)$ and $f_2(x) = x + x^{\sqrt{2}}$ are strictly increasing bijections of $(0, \infty)$ onto itself, with

$$f_1'(x) = 1 + x(3 + 2 \log x) \geq m_1 := 1 - 2e^{-5/2} > 0.835, \quad f_2'(x) = 1 + \sqrt{2}x^{\sqrt{2}-1} > 1. \quad (4.5)$$

Their inverses g_1, g_2 are C^1 up to $y = 0$ with $g_i(0) = 0$, $g_i'(0) = 1$, but g_1 is not C^2 at 0: $g_1''(y) = -f_1''(g_1)/f_1'(g_1)^3 \sim -(5 + 2 \log y) \rightarrow +\infty$.

Proof. $x(3 + 2 \log x)$ has derivative $5 + 2 \log x$, minimum at $x = e^{-5/2}$ with value $-2e^{-5/2}$; the limits of f_1 at 0 and ∞ are 0 and ∞ . The rest is immediate. $\square$

So for both examples the branch of Theorem 4.1 is the real inverse, and the expansions (1.3)–(1.4) describe it near 0; the absence of a second-order Taylor expansion of g_1 at 0 is exactly why $\log y$ must appear at y^2 .

4.3 The complex germ returned by InverseFunction

The equation $f_1(x) = 0$, $x \neq 0$, reads $1 + x(1 + \log x) = 0$. With $t = 1 + \log x$ it becomes $te^t = -e$, so $t = W_k(-e)$ and

$$c_k = e^{W_k(-e)-1} = -\frac{1}{W_k(-e)}, \quad f_1'(c_k) = c_k - 1, \quad f_1''(c_k) = 3 + 2W_k(-e), \quad (4.6)$$

where the derivative identities follow from $c_k W_k(-e) = -1$. Each c_k is a nonzero complex point where f_1 is analytic with nonzero derivative, so the analytic inverse function theorem gives an inverse analytic near $y = 0$ with value c_k at 0; `Series[InverseFunction[...][y], {y, 0, 2}]` computes exactly that germ for $k = -1$:

$$c + \frac{y}{c-1} - \frac{f_1''(c)}{2f_1'(c)^3}y^2 + \dots = c + \frac{y}{c-1} - \frac{3 + 2W_{-1}(-e)}{2(c-1)^3}y^2 + \mathcal{O}(y^3),$$

which is (1.2). Nothing is wrong with this answer: it is the inverse of a different branch of f_1^{-1} , the one through a complex root, and adding `Assumptions -> y > 0` cannot repair it, because the reality of y says nothing about which preimage is intended. The real branch is characterised by $g(y) \rightarrow 0$, and that condition, not an assumption on y , is what an inversion routine must be given; the package builds the branch of Theorem 4.1 directly and never asks `InverseFunction` to choose.

5 Lagrange–Bürmann inversion with a marker

The classical Lagrange inversion theorem expands the inverse of an analytic map at a point where its derivative does not vanish. Our maps are not analytic at 0, and the inverse has no Taylor series there; but at every *positive* point the map is analytic, and the trick is to invert a perturbation $u + \eta h(u) = q$ around the positive point q in the perturbation parameter η . The logarithms and irrational powers in h are then analytic functions of u near $q > 0$, and the derivatives that appear are computed by the Euler identity without ever evaluating anything at 0.

5.1 The marker theorem

Theorem 5.1 (Lagrange–Bürmann with a marker). *Let $q > 0$ and let h and Φ be analytic in a complex neighbourhood of q . For $|\eta|$ sufficiently small the equation*

$$u + \eta h(u) = q \quad (5.1)$$

has a unique solution $u(\eta)$ near q with $u(0) = q$, it is analytic in η , and

$$\Phi(u(\eta)) = \Phi(q) + \sum_{N \geq 1} \frac{(-\eta)^N}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi'(q) h(q)^N]. \quad (5.2)$$

The same identity holds coefficientwise in η for arbitrary formal power series h, Φ with coefficients in any commutative $\mathbb{Q}$ -algebra, i.e. as an identity of formal power series in η whose coefficients are polynomials in finitely many derivatives of h and Φ at q .

Proof. Choose $\rho > 0$ so small that h and Φ are analytic on the closed disc $|s - q| \leq \rho$, and let $M = \max_{|s-q|=\rho} |h(s)|$. For $|\eta| < \rho/M$ we have $|\eta h(s)| < |s - q|$ on the circle $C = \{|s - q| = \rho\}$, so by Rouché's theorem $F_\eta(s) = s - q + \eta h(s)$ has exactly one zero $u(\eta)$ in the disc, and by the argument principle

$$\Phi(u(\eta)) = \frac{1}{2\pi i} \oint_C \Phi(s) \frac{F'_\eta(s)}{F_\eta(s)} ds = \frac{1}{2\pi i} \oint_C \Phi(s) \frac{1 + \eta h'(s)}{s - q + \eta h(s)} ds.$$

On C the geometric series $\frac{1}{s - q + \eta h(s)} = \sum_{N \geq 0} \frac{(-\eta)^N h(s)^N}{(s - q)^{N+1}}$ converges uniformly, so

$$\Phi(u(\eta)) = \sum_{N \geq 0} (-\eta)^N \left[\frac{1}{2\pi i} \oint_C \frac{\Phi(s) h(s)^N}{(s - q)^{N+1}} ds - \frac{1}{2\pi i} \oint_C \frac{\Phi(s) h'(s) h(s)^N (-\eta)}{(s - q)^{N+1}} ds \right].$$

By Cauchy's formula the first integral is $\frac{1}{N!} \frac{d^N}{dq^N} [\Phi h^N](q)$ and the second, after shifting the index $N \rightarrow N - 1$, contributes $-\frac{1}{(N-1)!} \frac{d^{N-1}}{dq^{N-1}} [\Phi h' h^{N-1}](q)$ to the coefficient of $(-\eta)^N$ for $N \geq 1$. Since $\frac{d}{dq} [\Phi h^N] = \Phi' h^N + N \Phi h' h^{N-1}$, the coefficient of $(-\eta)^N$ is

$$\frac{1}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi' h^N + N \Phi h' h^{N-1}] - \frac{1}{(N-1)!} \frac{d^{N-1}}{dq^{N-1}} [\Phi h' h^{N-1}] = \frac{1}{N!} \frac{d^{N-1}}{dq^{N-1}} [\Phi' h^N],$$

which is (5.2); the constant term is $\Phi(q)$. Analyticity of $u(\eta)$ follows from the implicit function theorem at $(q, 0)$, where $\partial_u F_0 = 1$. For the formal statement, both sides of (5.2) have coefficients that are universal polynomials in the Taylor coefficients of h and Φ at q ; two polynomials that agree for all analytic data agree identically (they agree on a Zariski-dense set), so the identity holds for arbitrary formal coefficients. $\square$

Remark 5.2. Other proofs exist [4, 5, 3]; the one above is chosen because it exhibits the analytic content: the solution is the unique root in a disc around q , a fact reused in Section 7 to obtain explicit constants. The formal statement is what makes the coefficient formula of Section 6 an identity, independent of any convergence question.

5.2 Consequences

Near-identity form. With $\Phi(q) = q$ and $\eta = 1$,

$$f(u) = u + h(u) \implies g(y) = y + \sum_{N \geq 1} \frac{(-1)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} [h(y)^N], \quad (5.3)$$

formally; Sections 7 and 9 say when setting $\eta = 1$ is legitimate as an asymptotic statement. The formula (5.3) is a complete answer for the log example: with $h(y) = y^2(1 + \log y)$, the N -th term is $\frac{(-1)^N}{N!} \partial_y^{N-1} [y^{2N}(1 + \log y)^N]$, computed by Lemma 2.6; for $N = 2$, $\frac{1}{2} \partial_y [y^4(1 + L)^2] = y^3(2(1 + L)^2 + (1 + L)) = y^3(2L^2 + 5L + 3)$.

Several markers. If $h = \sum_i \eta_i h_i$ with independent markers, expanding h^N multinomially in (5.2) gives the coefficient of $\prod \eta_i^{k_i}$, $|\mathbf{k}| = N$, as

$$\frac{(-1)^N}{\mathbf{k}!} \frac{d^{N-1}}{dq^{N-1}} \left[\Phi'(q) \prod_i h_i(q)^{k_i} \right], \quad (5.4)$$

because the multinomial coefficient $N!/\mathbf{k}!$ cancels the $1/N!$.

Transport of observables. Formula (5.2) gives at once the expansion of any analytic function of the inverse. With $\Phi(u) = u^r$ one obtains powers of the inverse; with $\Phi = \log$ its logarithm; differentiating the identity for u^r with respect to r at $r = 0$ gives the logarithm again. These are used in Section 6.

General core. Suppose $F = F_0 + R$ where F_0 has a known inverse ϕ on the branch of interest and R is a perturbation. Put $t = F_0(x)$, so that $F(x) = y$ becomes $t + \eta R(\phi(t)) = y$ with $\eta = 1$; Theorem 5.1 with $\Phi = \phi$ yields

$$x = \phi(y) + \sum_{N \geq 1} \frac{(-1)^N}{N!} \frac{d^{N-1}}{dy^{N-1}} \left[\phi'(y) R(\phi(y))^N \right]. \quad (5.5)$$

This is the “phase-coordinate” inversion of [11]; it is implemented as `PerturbativeInverse` [ϕ , R , $\{\mathbf{x}, \mathbf{y}\}$, $\mathbf{n}$] and is the natural tool when the dominant part of F is not a monomial (Section 12). For the monomial core $F_0(x) = ax^p$, $\phi(y) = (y/a)^{1/p}$, it is equivalent to the substitution $t = x^p$ used in the next section.

6 The coefficient theorem

6.1 Statement

Let f be the germ (4.1), z the uniformizer (4.2), $L = \log z$, $\mathcal{D} = d/dL$. For a multi-index $\mathbf{k} \in \mathbb{N}^m$ write $n = |\mathbf{k}|$, $\sigma = \mathbf{k} \cdot \boldsymbol{\delta}$ and $B^{\mathbf{k}} = \prod_i B_i^{k_i}$.

Theorem 6.1 (Euler-operator coefficient formula). *For every real r , the branch g of Theorem 4.1 satisfies, formally and (Theorem 7.1) as a convergent expansion for small z ,*

$$g(y)^r = z^r + \sum_{\mathbf{k} \neq 0} z^{r+\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}^{(r)}(L), \quad C_{\mathbf{k}}^{(r)}(L) = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (\mathcal{D} + r + \sigma + pj) B^{\mathbf{k}}(L). \quad (6.1)$$

The empty product ($n = 1$) is the identity. Multi-indices of equal weight are summed: the block of g^r at $z^{r+\sigma}$ is $C_{\sigma}^{(r)} = \sum_{\mathbf{k} \cdot \boldsymbol{\delta} = \sigma} C_{\mathbf{k}}^{(r)}$. In particular

$$g(y) = z \left(1 + \sum_{\mathbf{k} \neq 0} z^{\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}(L) \right), \quad C_{\mathbf{k}} = C_{\mathbf{k}}^{(1)}, \quad (6.2)$$

$$\log g(y) = L + \sum_{\mathbf{k} \neq 0} z^{\mathbf{k} \cdot \boldsymbol{\delta}} \frac{(-1)^n}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (\mathcal{D} + \sigma + pj) B^{\mathbf{k}}(L).$$

Proof. Put $t = u^p$ and $q = v/a$ (so $q = z^p$). The equation $f(u) = y$ becomes

$$q = t + \sum_i t^{1+\delta_i/p} B_i\left(\frac{\log t}{p}\right),$$

which is (5.1) with markers: $t + \sum_i \eta_i h_i(t) = q$, $h_i(t) = t^{1+\delta_i/p} B_i(\log t/p)$, at $\eta_i = 1$. The observable is $\Phi(t) = t^{r/p} = u^r$. All h_i and Φ are analytic near any $q > 0$. By (5.4) the coefficient of $\prod \eta_i^{k_i}$ is

$$\frac{(-1)^n}{\mathbf{k}!} \frac{d^{n-1}}{dq^{n-1}} \left[\frac{r}{p} q^{r/p-1} \prod_i (q^{1+\delta_i/p} B_i(\frac{\log q}{p}))^{k_i} \right] = \frac{(-1)^n r}{p \mathbf{k}!} \frac{d^{n-1}}{dq^{n-1}} \left[q^{n-1+(r+\sigma)/p} B^{\mathbf{k}}(\frac{\log q}{p}) \right].$$

Apply the repeated Euler identity (2.2) in the variable q , with $\Lambda = \log q$ and the polynomial $\tilde{P}(\Lambda) = B^{\mathbf{k}}(\Lambda/p)$: the derivative of order $n-1$ of $q^b \tilde{P}(\Lambda)$, $b = n-1+(r+\sigma)/p$, is $q^{b-n+1} \prod_{j=0}^{n-2} (b-j + d/d\Lambda) \tilde{P}$. Since $L = \Lambda/p$, $d/d\Lambda = \mathcal{D}/p$, and $b-j = (r+\sigma+p(n-1-j))/p$; hence each factor is $\frac{1}{p}(\mathcal{D} + r + \sigma + p(n-1-j))$, and with $j' = n-1-j$ running from 1 to $n-1$ the product is $p^{-(n-1)} \prod_{j'=1}^{n-1} (\mathcal{D} + r + \sigma + pj') B^{\mathbf{k}}(L)$. Finally $q^{b-n+1} = q^{(r+\sigma)/p} = z^{r+\sigma}$, and the prefactor r/p combines with $p^{-(n-1)}$ into r/p^n . The formal validity is Theorem 5.1's formal clause, applied to the finitely many markers; setting all $\eta_i = 1$ is legitimate coefficientwise because by Lemma 2.4 only finitely many multi-indices contribute to any exponent. The formula for $\log g$ is the derivative of (6.1) with respect to r at $r = 0$, where the factor r vanishes and only the term with the factor differentiated survives; equivalently it is Theorem 5.1 with $\Phi = \log$. $\square$

Remark 6.2. The formula (6.1) is the one on which all nine reports of Section 14 agree, with the same normalization: sign $(-1)^n$, the multi-index factorial $\mathbf{k}!$ (not $n!$: the multinomial coefficient from expanding h^n has been absorbed), the factor p^{-n} (one $1/p$ from Φ' and one from each of the $n-1$ derivatives, through $d/d\Lambda = \mathcal{D}/p$), and the operator constants $r + \sigma + pj$. Forgetting the substitution $L = \log(v/a)/p$ in the coefficients, or the factor p^{-n} , are the two classical mistakes for nonunit p .

6.2 First orders and cross terms

For a single block ($m = 1$, gap δ , polynomial B):

$$g = z - \frac{z^{1+\delta}}{p} B + \frac{z^{1+2\delta}}{2p^2} ((1+2\delta+p)B^2 + 2BB') - \frac{z^{1+3\delta}}{6p^3} (\mathcal{D} + 1 + 3\delta + p)(\mathcal{D} + 1 + 3\delta + 2p)B^3 + \dots \quad (6.3)$$

For two blocks the mixed multi-index $e_i + e_j$ contributes

$$\frac{z^{1+\delta_i+\delta_j}}{p^2} (\mathcal{D} + 1 + \delta_i + \delta_j + p)(B_i B_j) = \frac{z^{1+\delta_i+\delta_j}}{p^2} ((1 + \delta_i + \delta_j + p)B_i B_j + B'_i B_j + B_i B'_j), \quad (6.4)$$

which shows why the inverse of a sum of perturbations is not the sum of the separately computed inverse corrections.

6.3 Degrees and leading logarithms

Proposition 6.3 (Degrees). *Let $d_i = \deg B_i$ and $D(\mathbf{k}) = \sum_i k_i d_i$. Then $\deg C_{\mathbf{k}}^{(r)} = D(\mathbf{k})$ exactly, with leading coefficient*

$$[L^{D(\mathbf{k})}] C_{\mathbf{k}}^{(r)} = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (r + \sigma + pj) \prod_i ([L^{d_i}] B_i)^{k_i}, \quad (6.5)$$

which is nonzero whenever $r + \sigma + pj \neq 0$ for $1 \leq j \leq n-1$ (always true for $r > 0, p > 0$). A block $C_\sigma^{(r)}$ of the aggregated expansion has degree at most $\max\{D(\mathbf{k}) : \mathbf{k} \cdot \boldsymbol{\delta} = \sigma\}$, with equality unless the leading coefficients of the contributing multi-indices cancel.

Proof. Each factor $\mathcal{D} + c$ with $c \neq 0$ preserves the degree and multiplies the top coefficient by c ; the product $B^{\mathbf{k}}$ has degree $D(\mathbf{k})$. $\square$

For the log example ($a = p = \delta = 1$, $B = 1 + L$, single index n) the formula reads

$$\begin{aligned} P_n(L) &= \frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} (\mathcal{D} + j) (1 + L)^n, \\ [L^n]P_n &= \frac{(-1)^n}{n!} \prod_{j=n+2}^{2n} j = (-1)^n \frac{(2n)!}{n!(n+1)!} = (-1)^n \text{Cat}_n, \end{aligned} \quad (6.6)$$

the signed Catalan numbers. The next coefficient is also explicit: selecting the derivative in exactly one factor of the product gives

$$[L^{n-1}]P_n = (-1)^n \text{Cat}_n n (1 + H_{2n} - H_{n+1}), \quad H_k = \sum_{j \leq k} 1/j, \quad (6.7)$$

because $\prod_{j=n+2}^{2n} (\mathcal{D} + j)(1 + L)^n$ has L^{n-1} -coefficient $(\prod j)(n + \sum_{j=n+2}^{2n} n/j)$ (the term n from expanding $(1 + L)^n$ and the terms n/j from applying $\mathcal{D}$ in the factor j). Every P_n is divisible by $1 + L$: in the variable $t = 1 + L$ the operator product $\prod_{j=n+2}^{2n} (\mathcal{D} + j)$ consists of $n-1$ factors, each of which either differentiates (lowering the power of t by one) or multiplies by a constant, so every monomial of $\prod (\mathcal{D} + j) t^n$ has degree at least $n - (n-1) = 1$ in t ; the package confirms the factorisation through $n = 8$. Thus

$$\begin{aligned} P_1 &= -(1 + L), & P_2 &= (1 + L)(2L + 3), & P_3 &= -\frac{1}{2}(1 + L)(10L^2 + 31L + 23), \\ P_4 &= \frac{1}{6}(1 + L)(84L^3 + 398L^2 + 607L + 299), \end{aligned}$$

which are the blocks of (1.3).

6.4 Constant coefficients: Fuss–Catalan numbers

When every B_i is a constant b_i , the operators act as scalars and

$$C_{\mathbf{k}}^{(r)} = \frac{(-1)^n r}{p^n \mathbf{k}!} \prod_{j=1}^{n-1} (r + \sigma + pj) \prod_i b_i^{k_i} = \frac{r}{r + \sigma} \binom{-(r + \sigma)/p}{n} \frac{n!}{\mathbf{k}!} \prod_i b_i^{k_i}. \quad (6.8)$$

For $f(x) = x + bx^\alpha$ ($p = 1$, $\delta = \alpha - 1$, $r = 1$) this is the generalised Fuss–Catalan law

$$\begin{aligned} g(y) &= \sum_{n \geq 0} c_n b^n y^{1+n(\alpha-1)}, \\ c_n &= \frac{(-1)^n}{n!} \prod_{j=0}^{n-2} (n\alpha - j) = \frac{(-1)^n}{1 + n(\alpha - 1)} \binom{n\alpha}{n} = (-1)^n \frac{\Gamma(n\alpha + 1)}{\Gamma(n + 1)\Gamma(n(\alpha - 1) + 2)}, \end{aligned} \quad (6.9)$$

valid for every real $\alpha > 1$ (no rationality is used). For $\alpha = 2$ the c_n are the signed Catalan numbers and $g = (\sqrt{1 + 4y} - 1)/2$; for $\alpha = \sqrt{2}$,

$$\begin{aligned} c_1 &= -1, & c_2 &= \sqrt{2}, & c_3 &= -\frac{6 - \sqrt{2}}{2}, \\ c_4 &= \frac{17\sqrt{2}}{3} - 4, & c_5 &= \frac{51\sqrt{2}}{4} - \frac{305}{12}, & c_6 &= \frac{322\sqrt{2}}{5} - 77, \end{aligned}$$

which gives (1.4). In this one-gap case $g(y) = yV(y^{\alpha-1})$ with V the solution of $V + tV^\alpha = 1$, $V(0) = 1$, analytic at $t = 0$ with radius of convergence

$$R_\alpha = \frac{(\alpha-1)^{\alpha-1}}{\alpha^\alpha}, \quad |c_n| \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}(\alpha-1)^{3/2}} n^{-3/2} R_\alpha^{-n}, \quad (6.10)$$

by Stirling's formula, so the expansion (1.4) converges (and equals the positive inverse, by analytic continuation of the implicit equation) for $0 < y < R_{\sqrt{2}}^{1/(\sqrt{2}-1)} \approx 0.12686$. The singularity at $t = -R_\alpha$ is the fold $V = \alpha/(\alpha-1)$ of the implicit equation on the negative axis.

6.5 Formal existence, uniqueness, and the two iterative engines

Theorem 6.1 produces the coefficients; the following independent argument shows that they are the only possible ones and provides two alternative algorithms. Write $g = z(1 + U)$ with U an unknown jet of positive valuation. Then $f(g) = y$ is equivalent to

$$E(U) := (1 + U)^p \left(1 + \sum_i z^{\delta_i} (1 + U)^{\delta_i} B_i(L + \log(1 + U)) \right) - 1 = 0, \quad (6.11)$$

an equation in the jet algebra (Lemma 2.7 evaluates every term).

Proposition 6.4 (Triangular structure). *Equation (6.11) has exactly one formal solution U of positive valuation with support in the semigroup S generated by the gaps. Its coefficient at z^σ is determined by the coefficients at smaller weights through one division by p .*

Proof. $E(U) = pU + \sum_i z^{\delta_i} B_i(L) + (\text{terms of weight } \geq \min(2 \text{ val } U, \text{ val } U + \delta_*))$, because $(1 + U)^p = 1 + pU + \mathcal{O}(U^2)$ and every correction carries a factor z^{δ_i} . Order the elements of S increasingly (Lemma 2.4); at a new weight σ the coefficient of z^σ in $E(U)$ is pU_σ plus an expression in the $U_{\sigma'}$ with $\sigma' < \sigma$, which fixes U_σ . If U, V are two solutions with $U - V$ of least weight σ , then $E(U) - E(V) = (U - V)(p + \text{positive weight})$ has a nonzero block at z^σ , a contradiction. $\square$

Picard iteration. Rewriting (6.11) as $U = \Psi(U) := (1 + \sum_i z^{\delta_i} (1 + U)^{\delta_i} B_i(L + \log(1 + U)))^{-1/p} - 1$, the map Ψ gains one gap per step: $\text{val}(\Psi(U) - \Psi(V)) \geq \text{val}(U - V) + \delta_*$. Hence $U_{k+1} = \Psi(U_k)$, $U_0 = 0$, agrees with the solution below weight $(k+1)\delta_*$, and $\lceil H/\delta_* \rceil$ iterations give all blocks below H .

Newton iteration. With $E'(U) = p(1 + U)^{p-1} + \sum_i z^{\delta_i} (1 + U)^{p+\delta_i-1} ((p + \delta_i)B_i + B'_i)(L + \log(1 + U))$, a unit of the jet algebra with constant leading coefficient p , the iteration $U_{k+1} = U_k - E(U_k)/E'(U_k)$ doubles the valuation of the error: if $\text{val}(U_k - U) \geq s$ then $\text{val}(U_{k+1} - U) \geq 2s$, because $E(U_k) = E'(U)(U_k - U) + \mathcal{O}((U_k - U)^2)$ and $E'(U_k)^{-1} = E'(U)^{-1}(1 + \mathcal{O}(U_k - U))$. Starting from $U_0 = 0$ with $s = \delta_*$, about $\log_2(H/\delta_*)$ iterations suffice. The package offers this engine as `Method -> "Newton"`; it and the multi-index formula must agree, which is a strong internal check.

7 Convergence

A formal identity such as (6.1) says nothing, by itself, about the function g : the series might diverge, or converge to something else. This section proves that the expansion converges absolutely for small z and equals the branch of Theorem 4.1, first by an implicit-function argument that also yields the coefficient bounds needed for the remainder theory, then by a Rouché argument that yields explicit constants.

7.1 Convergence in finitely many small parameters

Theorem 7.1 (Convergence). *Let f be the germ (4.1) and g its branch. For every real r the series (6.1) converges absolutely for all sufficiently small $z > 0$, its sum is $g(y)^r$, and there are constants M, K such that*

$$|C_{\mathbf{k}}^{(r)}(L)| \leq M K^{|\mathbf{k}|} M(z)^{D(\mathbf{k})} \quad (L \leq \log z_1), \quad D(\mathbf{k}) = \sum_i k_i \deg B_i. \quad (7.1)$$

In particular $g^r \in \mathcal{E}$ (Definition 3.1) with support $r + S$.

Proof. Write $g = zw$ with $w \rightarrow 1$; by (6.11) the unknown w solves

$$w^p \left(1 + \sum_i z^{\delta_i} w^{\delta_i} B_i(L + \log w) \right) = 1.$$

Expand each $B_i(L + \log w) = \sum_{j=0}^{d_i} L^j A_{ij}(\log w)$ with $A_{ij}(v) = \sum_{s \geq j} b_{is} \binom{s}{j} v^{s-j}$, where $B_i = \sum_s b_{is} L^s$. Introduce independent complex parameters T_{ij} in place of $z^{\delta_i} L^j$ and consider

$$\mathcal{E}(w, T) = w^p \left(1 + \sum_{i,j} T_{ij} w^{\delta_i} A_{ij}(\log w) \right) - 1. \quad (7.2)$$

On the disc $|w - 1| < 1$ with the principal logarithm, $\mathcal{E}$ is analytic in (w, T) ; $\mathcal{E}(1, 0) = 0$ and $\partial_w \mathcal{E}(1, 0) = p \neq 0$. The analytic implicit function theorem gives an analytic $W(T)$ on a polydisc $|T_{ij}| < \rho$ with $W(0) = 1$ and $\mathcal{E}(W(T), T) = 0$, and W^r is analytic there too. Its Taylor series $W(T)^r = \sum_{\nu} c_{\nu} T^{\nu}$ converges absolutely on the smaller polydisc $|T_{ij}| \leq \rho/2$, with the Cauchy estimates $|c_{\nu}| \leq M(2/\rho)^{|\nu|}$.

Now specialise $T_{ij} = z^{\delta_i} L^j$. Since $z^{\delta_i} M(z)^{d_i} \rightarrow 0$, the point lies in the polydisc for small z , and $W(T(z))$ is real (all data are real and the solution is unique) and close to 1, so $zW(T(z))$ is a solution of $f(u) = y$ with $u \sim z$; by Theorem 4.1 it is $g(y)$. Group the absolutely convergent series $\sum_{\nu} c_{\nu} \prod_{i,j} (z^{\delta_i} L^j)^{\nu_{ij}}$ by the multi-index $k_i = \sum_j \nu_{ij}$: the terms with a given $\mathbf{k}$ are finitely many (at most $\prod_i (d_i + 1)^{k_i}$), and their sum is $z^{\mathbf{k} \cdot \delta}$ times a polynomial in L of degree at most $\sum_j j \nu_{ij} \leq D(\mathbf{k})$. Absolute convergence permits this regrouping, so $g^r = z^r \sum_{\mathbf{k}} z^{\mathbf{k} \cdot \delta} \tilde{C}_{\mathbf{k}}(L)$ with the bound (7.1) for $K = (2/\rho) \max_i (d_i + 1)$. Finally the $\tilde{C}_{\mathbf{k}}$ are the coefficients $C_{\mathbf{k}}^{(r)}$ of (6.1): both are formal solutions of the same triangular problem (Proposition 6.4 for $r = 1$; for general r apply the observable $w \mapsto w^r$), which has a unique solution. $\square$

Remark 7.2. The theorem does not give a radius of convergence in y : logarithms are present, and the expansion is not a power series in any single variable (except in the one-gap constant-coefficient case, (6.10)). It gives convergence of a multivariate power series after the substitution $T_{ij} = z^{\delta_i} (\log z)^j$, uniformly on $0 < z < z_1$. Nor is anything uniform in the parameters a, p, δ_i, B_i : as a gap $\delta \rightarrow 0$ the hierarchy collapses, and the expansion of $x + x^\alpha$ at $\alpha \downarrow 1$ has radius (6.10) tending to 0.

7.2 An explicit conditional majorant

The implicit-function argument has no explicit constants. Rouché's theorem in the marker equation gives them, at the price of a hypothesis that must be checked at each y .

Theorem 7.3 (Explicit tail bound). *Consider the near-identity germ $f(t) = t + h(t)$, $h(t) = \sum_i t^{1+\alpha_i} P_i(\log t)$ with $\alpha_i > 0$ and $P_i(L) = \sum_s p_{is} L^s$ (the general germ reduces to this form by $t = u^p$, $q = v/a$). Fix $0 < r < 1$, put $c_r = -\log(1 - r)$ and*

$$\mathcal{Q}_r(t) = \frac{1}{r} \sum_i t^{\alpha_i} (1 + r)^{1+\alpha_i} \sum_s |p_{is}| (|\log t| + c_r)^s. \quad (7.3)$$

Let $t > 0$ satisfy $\mathcal{Q}_r(t) < 1$. Then:

1. for every complex λ with $|\lambda| < 1/\mathcal{Q}_r(t)$ the equation $u + \lambda h(u) = t$ has exactly one solution $u(\lambda)$ in the disc $|u - t| < rt$; it is analytic in λ , $u(0) = t$, and $u(\lambda)$ is real for real λ ;
2. the Lagrange terms $u_N = \frac{(-1)^N}{N!} \frac{d^{N-1}}{dt^{N-1}} [h(t)^N]$ satisfy $|u_N| \leq \frac{rt}{N} \mathcal{Q}_r(t)^N$, and $u(\lambda) = t + \sum_{N \geq 1} u_N \lambda^N$ converges absolutely for $|\lambda| < 1/\mathcal{Q}_r(t)$, in particular at $\lambda = 1$;
3. $u(1)$ is the positive solution of $f(u) = t$ in $(t(1-r), t(1+r))$, it coincides with the branch $g(t)$ for all sufficiently small t , and

$$\left| u(1) - t - \sum_{N=1}^{N_0} u_N \right| \leq \frac{rt}{N_0+1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}. \quad (7.4)$$

Proof. On the circle $|s - t| = rt$ write $s = t(1 + \zeta)$ with $|\zeta| = r < 1$. Then $|s| \leq (1+r)t$ and, for the principal logarithm, $|\log s| \leq |\log t| + |\log(1 + \zeta)| \leq |\log t| + \sum_{k \geq 1} r^k/k = |\log t| + c_r$. Hence $|h(s)| \leq \sum_i ((1+r)t)^{1+\alpha_i} \sum_s |p_{is}| (|\log t| + c_r)^s = rt \mathcal{Q}_r(t)$. For $|\lambda| \mathcal{Q}_r(t) < 1$ this gives $|\lambda h(s)| < rt = |s - t|$ on the circle, and Rouché's theorem applied to $s - t + \lambda h(s)$ and $s - t$ shows that the former has exactly one zero in the disc, which is (1); analyticity in λ follows from the argument-principle integral as in Theorem 5.1, and reality from uniqueness together with the invariance of the equation under conjugation. The coefficients u_N of the analytic function $\lambda \mapsto u(\lambda) - t$ are given by Theorem 5.1; Cauchy's estimate for the $(N-1)$ -st derivative of h^N on the circle of radius rt gives $|u_N| \leq \frac{1}{N!} \frac{(N-1)! (rt \mathcal{Q}_r(t))^N}{(rt)^{N-1}} = \frac{rt}{N} \mathcal{Q}_r(t)^N$, which is (2). For (3), $u(1)$ is real and in the disc, hence in $(t(1-r), t(1+r))$; $g(t) \sim t$ lies in that interval for small t and is also a root, so uniqueness gives $u(1) = g(t)$; and the tail bound is $\sum_{N > N_0} \frac{rt}{N} \mathcal{Q}_r(t)^N \leq \frac{rt}{N_0+1} \sum_{N > N_0} \mathcal{Q}_r(t)^N$. $\square$

Corollary 7.4 (Exponent cutoff). *Let G_B retain all multi-indices of weight $\mathbf{k} \cdot \boldsymbol{\alpha} \leq B$. Every multi-index with $|\mathbf{k}| \leq N_0 := \lfloor B/\alpha_{\max} \rfloor$ is retained, so under $\mathcal{Q}_r(t) < 1$*

$$|g(t) - G_B(t)| \leq \frac{rt}{N_0+1} \frac{\mathcal{Q}_r(t)^{N_0+1}}{1 - \mathcal{Q}_r(t)}.$$

Proof. The omitted part of the multi-marker expansion is a subseries of $\sum_{N > N_0} u_N$ (the expansion of u_N into multi-indices is absolutely convergent by the same Cauchy estimates, since $\sum_{|\mathbf{k}|=N} \frac{\prod M_i^{k_i}}{\mathbf{k}!} = \frac{(\sum M_i)^N}{N!}$). $\square$

Example 7.5. For $f_1(x) = x + x^2(1 + \log x)$ and $r = \frac{1}{2}$: $\mathcal{Q}_{1/2}(y) = \frac{9}{2} y (1 + |\log y| + \log 2)$, so for $y \leq \frac{1}{100}$ (where $\mathcal{Q}_{1/2} \leq 0.284$) the expansion converges and the error after N_0 blocks is at most $\frac{y}{2(N_0+1)} \mathcal{Q}_{1/2}(y)^{N_0+1}/(1 - \mathcal{Q}_{1/2}(y))$; at $y = 10^{-2}$ and $N_0 = 3$ this is $1.1 \cdot 10^{-5}$ while the actual error is $1.5 \cdot 10^{-7}$. The bound is conditional (it says nothing when $\mathcal{Q}_r \geq 1$) and not tight, but it is rigorous, cheap, and turns the asymptotic statements into inequalities at specific points.

8 Remainders after a cutoff

8.1 Exponent cutoff: the first omitted block

Let the target cutoff be an exponent q of the local variable of the result, exclusive. In terms of weights, with r the observable power, a block $z^{r+\sigma} C(L)$ has exponent $(r + \sigma)/|p|$ in the local variable of y (Section 10), so the retained set is

$$I_H = \{\mathbf{k} \in \mathbb{N}^m : \mathbf{k} \cdot \boldsymbol{\delta} < H\}, \quad H = |p|q - r, \quad (8.1)$$

a finite downward-closed set (Lemma 2.4). Its *one-step boundary* is

$$\partial I_H = \{\mathbf{k} + e_i : \mathbf{k} \in I_H, 1 \leq i \leq m\} \setminus I_H, \quad (8.2)$$

also finite. Put

$$\sigma_* = \min\{\boldsymbol{\nu} \cdot \boldsymbol{\delta} : \boldsymbol{\nu} \in \partial I_H\}, \quad C_{\sigma_*}^{(r)} = \sum_{\mathbf{k} \cdot \boldsymbol{\delta} = \sigma_*} C_{\mathbf{k}}^{(r)}, \quad k_* = \deg C_{\sigma_*}^{(r)} \quad (k_* = 0 \text{ if } C_{\sigma_*}^{(r)} = 0). \quad (8.3)$$

Lemma 8.1. $\sigma_* = \min(S \setminus [0, H))$, and every multi-index of weight σ_* lies in ∂I_H . Thus the complete coefficient at the first omitted weight is obtained from the boundary alone.

Proof. Let $\boldsymbol{\nu} \notin I_H$ have weight σ . Along a path from 0 to $\boldsymbol{\nu}$ that increases one coordinate at a time, the weights increase strictly, and the first point outside I_H is a boundary point of weight $\leq \sigma$; hence $\sigma_* \leq \sigma$, so σ_* is the least weight outside $[0, H)$. If $\sigma = \sigma_*$ the first point outside I_H has weight exactly σ_* by minimality, and no later point on the path can have the same weight, so $\boldsymbol{\nu}$ is that boundary point. $\square$

Theorem 8.2 (Remainder after an exponent cutoff). Let $G_H^{(r)} = z^r \sum_{\mathbf{k} \in I_H} z^{\mathbf{k} \cdot \boldsymbol{\delta}} C_{\mathbf{k}}^{(r)}(L)$. Then, as $z \rightarrow 0^+$,

$$g(y)^r - G_H^{(r)}(y) = z^{r+\sigma_*} C_{\sigma_*}^{(r)}(L) + o(z^{r+\sigma_*}) = \mathcal{O}(z^{r+\sigma_*} M(z)^{k_*}). \quad (8.4)$$

If $C_{\sigma_*}^{(r)} \neq 0$ the bound is attained by its top logarithm; if $C_{\sigma_*}^{(r)} = 0$ (a resonant cancellation) the remainder is $o(z^{r+\sigma_*})$, and the true order is that of the next nonzero block, which the theorem does not identify.

Proof. By Theorem 7.1, $g^r \in \mathcal{E}$ with the bound (7.1), and the omitted multi-indices are those of weight $\geq \sigma_*$ by Lemma 8.1; Lemma 3.2 (refined form) gives the claim. $\square$

Corollary 8.3 (In the variable of the result). With w the local variable of y ($w = |y - y_0|$ for finite y_0 , $w = 1/|y|$ for infinite y_0),

$$g(y)^r - G_H^{(r)}(y) = \mathcal{O}(w^{(r+\sigma_*)/|p|} M(w)^{k_*}), \quad \frac{r + \sigma_*}{|p|} \geq q,$$

since $z = (w/|a|)^{1/|p|}$ and $M(z) \asymp M(w)$.

The package reports the pair $((r + \sigma_*)/|p|, k_*)$ as `PowerLogRemainder[w,  $\beta$ , k]` and the block $z^{r+\sigma_*} C_{\sigma_*}^{(r)}$ as `"FrontierTerm"`. When the frontier coefficient cancels, the reported class is still valid, but a request for the next terms is the only way to learn the true order.

Example 8.4 (Resonances). For $f = x + x^2 + x^3$ the gaps are 1 and 2; at weight 2 the indices (2, 0) and (0, 1) contribute 2 and -1 , at weight 3 the indices (3, 0) and (1, 1) contribute -5 and $+5$, so

$$g(y) = y - y^2 + y^3 + 0 \cdot y^4 - 4y^5 + 14y^6 - 30y^7 + 33y^8 + \dots,$$

and after the cutoff 4 the reported remainder is $\mathcal{O}(y^4)$ while the truth is $\mathcal{O}(y^5)$. For $f = x + x^2 + 2x^3$ the block at y^3 cancels instead. Exact arithmetic on the exponents is what makes such cancellations detectable at all: with floating-point exponents, (2, 0) and (0, 1) might not be recognised as the same weight.

Example 8.5 (The log example). Here $m = 1$, $\delta = 1$, $B = 1 + L$, so $I_H = \{0, \dots, N\}$ for $H = N + 1$, the boundary is $\{N + 1\}$, and by (6.6) the frontier block $y^{N+2}P_{N+1}(\log y)$ has degree exactly $N + 1$ with top coefficient $(-1)^{N+1} \text{Cat}_{N+1}$. Hence

$$g_1(y) = y + \sum_{n=1}^N y^{n+1} P_n(\log y) + \mathcal{O}(y^{N+2} M(y)^{N+1}), \quad (8.5)$$

$$\frac{g_1(y) - y - \sum_{n \leq N} y^{n+1} P_n(\log y)}{y^{N+2} (\log y)^{N+1}} \rightarrow (-1)^{N+1} \text{Cat}_{N+1},$$

and in particular after $y - y^2(1 + \log y)$ the error is $y^3(2 \log^2 y + 5 \log y + 3) + \dots$, not $\mathcal{O}(y^3)$. The ratio in (8.5) converges slowly (like $1 + c/\log y$): for $N = 1$ it is 1.89 at $y = 10^{-20}$ and 1.98 at $y = 10^{-100}$.

8.2 Depth truncation

When the exponents are symbolic (Section 12) their order may be undecidable, but the total degree $|\mathbf{k}|$ of a multi-index is always known.

Theorem 8.6 (Remainder after a depth cutoff). *Let $G_{\leq N}^{(r)} = z^r \sum_{|\mathbf{k}| \leq N} z^{\mathbf{k} \cdot \delta} C_{\mathbf{k}}^{(r)}(L)$, $\delta_* = \min_i \delta_i$ and $d_* = \max_i \deg B_i$. Then*

$$g(y)^r - G_{\leq N}^{(r)}(y) = \mathcal{O}(z^{r+(N+1)\delta_*} M(z)^{(N+1)d_*}). \quad (8.6)$$

Proof. The omitted terms have $|\mathbf{k}| \geq N + 1$; use (7.1) and the count m^n of multi-indices of degree n exactly as in the proof of Lemma 3.2. $\square$

Depth truncation retains every block of exponent below $r + (N + 1)\delta_*$, but it may also retain blocks above that bound while omitting smaller ones when several gaps are present: for $f = x + x^2 + x^4$ depth 1 gives $y - y^2 - y^4$ although the omitted depth-two term $2y^3$ dominates y^4 . The remainder (8.6) is nevertheless correct as stated.

8.3 Numerical illustration

Table 1 shows, for the two examples of [1] and for the inverse of (1.5) at infinity, the actual error $|g(y) - G(y)|$ of the truncated expansion (the exact inverse computed by `FindRoot` at 80 digits) and its ratio to the reported remainder scale $w^{\beta_*} M(w)^{k_*}$. The ratios stay bounded, and in the log example they grow slowly with N because the leading coefficient of the frontier block, Cat_{N+1} , does.

9 Transport of remainders and finite smoothness

The theory so far concerns an exact finite germ. In practice the forward function is often known only through a truncated expansion, and one wants to know how the omitted part affects the inverse; conversely, one wants to turn a small residual $f(G(y)) - y$ of an approximation G into a bound for $G - g$. Both are mean-value arguments, but the scaling factor z^{1-p} of the derivative must not be forgotten.

$f_1 = x + x^2(1 + \log x)$, N corrections	$y = 10^{-2}$		$y = 10^{-4}$		$y = 10^{-8}$	
	error	ratio	error	ratio	error	ratio
$N = 1$	$2.4 \cdot 10^{-5}$	0.77	$1.3 \cdot 10^{-10}$	1.22	$5.9 \cdot 10^{-22}$	1.56
$N = 2$	$1.8 \cdot 10^{-6}$	1.03	$2.4 \cdot 10^{-13}$	2.26	$2.5 \cdot 10^{-28}$	3.38
$N = 3$	$1.5 \cdot 10^{-7}$	1.51	$5.1 \cdot 10^{-16}$	4.69	$1.2 \cdot 10^{-34}$	8.18
$N = 5$	$1.2 \cdot 10^{-9}$	3.75	$2.7 \cdot 10^{-21}$	24.0	$3.1 \cdot 10^{-47}$	57.4
$f_2 = x + x^{\sqrt{2}}$, N corrections						
$N = 1$	$2.5 \cdot 10^{-4}$	1.14	$6.6 \cdot 10^{-8}$	1.37	$3.3 \cdot 10^{-15}$	1.41
$N = 2$	$6.0 \cdot 10^{-5}$	1.82	$2.4 \cdot 10^{-9}$	2.21	$2.6 \cdot 10^{-18}$	2.29
$N = 3$	$1.5 \cdot 10^{-5}$	3.16	$9.1 \cdot 10^{-11}$	3.86	$2.2 \cdot 10^{-21}$	4.01
$N = 5$	$1.2 \cdot 10^{-6}$	10.9	$1.5 \cdot 10^{-13}$	13.5	$1.8 \cdot 10^{-27}$	14.1
inverse of $(1 + x + x^{\sqrt{2}})^{\sqrt{2}}$, n blocks						
	$y = 10^4$		$y = 10^8$		$y = 10^{16}$	
$n = 2$	0.72	0.33	1.9	0.39	9.3	0.40
$n = 4$	$9.6 \cdot 10^{-2}$	0.65	$1.5 \cdot 10^{-2}$	0.69	$3.4 \cdot 10^{-4}$	0.71
$n = 7$	$2.8 \cdot 10^{-4}$	0.039	$1.2 \cdot 10^{-6}$	0.023	$4.0 \cdot 10^{-11}$	0.015

Table 1: Errors of truncated expansions and their ratio to the remainder scale reported by the package. In the third group the local variable is $1/y$ and the leading term of the inverse is $\sqrt{y}$, so the absolute errors grow while the relative errors decrease.

9.1 From the forward function to the inverse

Theorem 9.1 (Transport of a forward remainder). *Let f be the germ (4.1) with $p > 0$ and let $F = f + R$ on $(0, u_0)$, where*

$$R(u) = \mathcal{O}(u^\rho M(u)^k), \quad R'(u) = \mathcal{O}(u^{\rho-1} M(u)^k), \quad \rho > p, \quad k \in \mathbb{N}. \quad (9.1)$$

Then F has a local inverse G on the same branch as g , and

$$G(y) - g(y) = \mathcal{O}(z^{\rho-p+1} M(z)^k) = \mathcal{O}(w^{(\rho-p+1)/p} M(w)^k), \quad w = |y - y_0|. \quad (9.2)$$

The same conclusion holds if the derivative hypothesis is replaced by the assumption that F is monotone near 0.

Proof. From (9.1) and Lemma 2.2, $F(u) - y_0 \sim au^p$ and $F'(u) \sim apu^{p-1}$, so F is strictly monotone near 0 and its inverse G satisfies $G \sim z$ (Theorem 4.1 applies verbatim). Put $u = G(y)$, $v = g(y)$; both are $\sim z$. Then $F(v) - y = F(v) - f(v) = R(v) = \mathcal{O}(z^\rho M(z)^k)$, while $F(u) - y = 0$. On the interval between u and v , which lies in $[z/2, 2z]$ for small z , F' is $\geq \frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$ in absolute value, so the mean value theorem gives $|u - v| \leq |R(v)|/\min |F'| = \mathcal{O}(z^{\rho-p+1} M(z)^k)$. Under monotonicity instead of the derivative bound, use f 's derivative: $f(u) - f(v) = y - R(u) - y = -R(u)$, and $f' \asymp z^{p-1}$ on $[z/2, 2z]$ gives the same estimate. $\square$

Remark 9.2. The derivative hypothesis is not a consequence of the value bound: $R(u) = u^2 \sin(u^{-2})$ is $\mathcal{O}(u^2)$, but $R'(u) = 2u \sin(u^{-2}) - 2u^{-1} \cos(u^{-2})$ destroys the monotonicity of $u + R(u)$, and $u + R$ has no inverse near 0. Truncated expansions produced by **Series** of log-exp functions do satisfy (9.1), because termwise differentiation of their asymptotic expansions is valid; the package records the hypothesis as "InputRemainder" $\rightarrow \{\rho, k\}$ and never proves it.

Corollary 9.3 (Cutoff cap). *If the inverse of F is computed from the truncated germ f with an exponent cutoff q in w , the result is a valid expansion of G only for $q \leq (\rho - p + 1)/p$; for such q the remainder class is the coarser of the truncation remainder (Corollary 8.3) and (9.2), compared by exponent first and by logarithmic degree at equal exponents.*

For example $\sin x + x^2 \log x = x + x^2 \log x - x^3/6 + \mathcal{O}(x^5)$ is a germ with $p = 1$, gaps 1, 2 and $\rho = 5$, $k = 0$; the inverse can be requested up to the cutoff $q = 5$ and is $y - y^2 \log y + y^3(\frac{1}{6} + \log y + 2 \log^2 y) + \mathcal{O}(y^4 M^3)$ for $q = 4$.

9.2 From a residual to the error of the inverse

Theorem 9.4 (Residual bracket). *Let F be continuous on $[A, B]$ and differentiable inside with $F' \geq \mu > 0$. Let $x_0 \in (A, B)$, $y \in \mathbb{R}$, $\varrho = F(x_0) - y$. If $[x_0 - |\varrho|/\mu, x_0 + |\varrho|/\mu] \subseteq [A, B]$, then $F(x) = y$ has exactly one solution x_* in $[A, B]$, it lies in that interval, and*

$$\frac{|\varrho|}{\max F'} \leq |x_* - x_0| \leq \frac{|\varrho|}{\mu}. \quad (9.3)$$

Proof. $F(x_0 + |\varrho|/\mu) \geq F(x_0) + |\varrho| \geq y$ and $F(x_0 - |\varrho|/\mu) \leq y$; the intermediate value theorem gives a root in the interval, monotonicity its uniqueness in $[A, B]$, and the mean value theorem both inequalities. $\square$

This gives an a posteriori certificate that needs no expansion theory, only a lower bound on the derivative. For the two examples the global bounds (4.5) give, for every $y > 0$ and every $G > 0$,

$$|G - g_1(y)| \leq \frac{|G + G^2(1 + \log G) - y|}{1 - 2e^{-5/2}}, \quad |G - g_2(y)| \leq |G + G^{\sqrt{2}} - y|. \quad (9.4)$$

For a general germ the derivative is $\asymp z^{p-1}$, and the bracket must be applied on $[z/2, 2z]$ with $\mu = \frac{1}{2}|ap|z^{p-1}2^{-|p-1|}$ (valid for small z), giving $|G - g| \leq \frac{2^{1+|p-1|}}{|ap|}z^{1-p}|f(G) - y|$.

The residual of the truncated expansion. Combining Theorem 8.2 with the mean value theorem, the truncated inverse G_H satisfies

$$\frac{f(G_H(y))}{a z^p} - 1 = -p z^{\sigma_*} C_{\sigma_*}(L) + o(z^{\sigma_*}), \quad (9.5)$$

since $f(G_H) - y \approx f'(g)(G_H - g) = -apz^{p-1} \cdot z^{1+\sigma_*} C_{\sigma_*}$. The normalized residual has the same first block as the error, up to the factor $-p$; the package's `InverseResidual` computes it exactly in the jet algebra (with a higher cutoff it displays the first omitted block; below the cutoff it must vanish identically, which is an independent check of every coefficient), and its “forward residual cutoff” in the variable of y is $q + (p - 1)/p$.

9.3 Finite smoothness suffices

The finite germ is analytic away from 0, and the convergence proof used that. For perturbations outside the finite power-log class (a coefficient $\sin(\log x)$, a term $e^{-1/x}$, a function known only through derivative bounds) the marker formula (5.3) can still be applied, and the following theorem, which appears in report 05 with a sketched proof, shows that it is a genuine asymptotic expansion whenever the perturbation and finitely many of its derivatives are small in the power-log scale. We give the complete proof.

Theorem 9.5 (Tame near-identity inversion). *Let $N \geq 0$, $\delta > 0$ and $M \geq 0$. Let φ be real-valued and C^{N+1} on $(0, x_0)$ with*

$$\varphi^{(j)}(x) = \mathcal{O}(x^{1+\delta-j} M(x)^M) \quad (x \rightarrow 0^+), \quad j = 0, 1, \dots, N + 1. \quad (9.6)$$

Then $f = x + \varphi$ has a unique local inverse g on $(0, y_0)$ with $g(y) \sim y$, and

$$g(y) = y + \sum_{n=1}^N \frac{(-1)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} [\varphi(y)^n] + \mathcal{O}(y^{1+(N+1)\delta} M(y)^{(N+1)M}). \quad (9.7)$$

Proof. Step 1: the homotopy. For $\lambda \in [0, 1]$ put $F_\lambda(x) = x + \lambda\varphi(x)$. By (9.6) with $j = 1$, $\varphi'(x) \rightarrow 0$, so there is x_1 with $F'_\lambda \geq \frac{1}{2}$ on $(0, x_1)$ for all $\lambda \in [0, 1]$; each F_λ is an increasing bijection of $(0, x_1)$ onto an interval containing $(0, y_1)$ for some y_1 independent of λ . Let $G(\lambda, y)$ be its inverse, so that

$$G + \lambda\varphi(G) = y, \quad G(0, y) = y. \quad (9.8)$$

Since $|\varphi(x)| \leq x/2$ for $x < x_1$ (after shrinking x_1), $F_\lambda(y/2) < y < F_\lambda(2y)$ and therefore $G(\lambda, y) \in [y/2, 2y]$ for all λ and all small y . The map $(\lambda, x) \mapsto x + \lambda\varphi(x) - y$ is C^{N+1} with x -derivative $\geq \frac{1}{2}$, so by the implicit function theorem G is C^{N+1} in (λ, y) , with

$$\partial_y G = \frac{1}{1 + \lambda\varphi'(G)} \in [\frac{2}{3}, 2], \quad \partial_\lambda G = -\frac{\varphi(G)}{1 + \lambda\varphi'(G)} = \psi(G) \partial_y G, \quad \psi := -\varphi. \quad (9.9)$$

Step 2: the transport identity. For every C^n function Ψ and $1 \leq n \leq N+1$,

$$\partial_\lambda^n [\Psi(G)] = \partial_y^{n-1} [\psi(G)^n \partial_y \Psi(G)]. \quad (9.10)$$

For $n = 1$: $\partial_\lambda \Psi(G) = \Psi'(G) \partial_\lambda G = \Psi'(G) \psi(G) \partial_y G = \psi(G) \partial_y [\Psi(G)]$. Assume (9.10) for n and all Ψ . Then, using it once more for the functions ψ^n and Ψ in the second step,

$$\begin{aligned} \partial_\lambda^{n+1} [\Psi(G)] &= \partial_y^{n-1} \partial_\lambda [\psi(G)^n \partial_y \Psi(G)] = \partial_y^{n-1} [\psi(G) \partial_y (\psi(G)^n \partial_y \Psi(G)) + \psi(G)^n \partial_y (\psi(G) \partial_y \Psi(G))] \\ &= \partial_y^{n-1} \partial_y [\psi(G)^{n+1} \partial_y \Psi(G)], \end{aligned}$$

where the last equality is the product rule read backwards: $\partial_y [\psi^{n+1} \Psi_y] = (n+1) \psi^n \psi_y \Psi_y + \psi^{n+1} \Psi_{yy}$ and the bracket equals $n \psi^n \psi_y \Psi_y + \psi^n (\psi_y \Psi_y + \psi \Psi_{yy})$, the same thing. (Here ψ_y stands for $\partial_y [\psi(G)]$.)

Step 3: the coefficients. At $\lambda = 0$, $G = y$, so (9.10) with $\Psi = \text{id}$ gives $\partial_\lambda^n G(0, y) = \partial_y^{n-1} [\psi(y)^n] = (-1)^n \partial_y^{n-1} [\varphi(y)^n]$, the Lagrange terms. Taylor's formula in λ with integral remainder gives

$$g(y) = G(1, y) = y + \sum_{n=1}^N \frac{(-1)^n}{n!} \partial_y^{n-1} [\varphi^n] + \frac{1}{N!} \int_0^1 (1-\lambda)^N \partial_\lambda^{N+1} G(\lambda, y) d\lambda.$$

Step 4: the remainder. Put $\kappa(y) = y^\delta M(y)^M$. Since $G \in [y/2, 2y]$ and $M(G) \leq 2M(y)$, hypothesis (9.6) gives $\varphi^{(j)}(G) = \mathcal{O}(\kappa y^{1-j})$ uniformly in λ , for $j \leq N+1$. We claim that for $1 \leq j \leq N+1$,

$$\partial_y^j G = \mathcal{O}(y^{1-j}) \quad (j \geq 1), \quad \partial_y^j [\varphi(G)] = \mathcal{O}(\kappa y^{1-j}) \quad (j \geq 0), \quad (9.11)$$

uniformly in $\lambda \in [0, 1]$. Differentiating (9.8) j times in y and solving for $\partial_y^j G$: $(1 + \lambda\varphi'(G)) \partial_y^j G$ equals δ_{j1} minus λ times a sum of terms $\varphi^{(m)}(G) \prod \partial_y^{i_s} G$ with $2 \leq m$, $\sum i_s = j$, $i_s \geq 1$ (Faà di Bruno). By induction on j each such term is $\mathcal{O}(\kappa y^{1-m}) \prod \mathcal{O}(y^{1-i_s}) = \mathcal{O}(\kappa y^{1-j})$, so $\partial_y^j G = \mathcal{O}(\kappa y^{1-j}) \subseteq \mathcal{O}(y^{1-j})$ for $j \geq 2$ (and $\partial_y G = \mathcal{O}(1)$). The same Faà di Bruno expansion with $m \geq 1$ gives the second claim. By Leibniz, $\partial_y^j [\psi(G)^{N+1}] = \mathcal{O}(\kappa^{N+1} y^{N+1-j})$, and $\partial_y^{N-j} [\partial_y G] = \partial_y^{N-j+1} G = \mathcal{O}(y^{-(N-j)})$. Applying (9.10) with $\Psi = \text{id}$ and $n = N+1$,

$$\partial_\lambda^{N+1} G = \partial_y^N [\psi(G)^{N+1} \partial_y G] = \sum_{j=0}^N \binom{N}{j} \partial_y^j [\psi(G)^{N+1}] \partial_y^{N-j+1} G = \mathcal{O}(\kappa^{N+1} y)$$

uniformly in λ , and the integral remainder is $\mathcal{O}(y \kappa(y)^{N+1}) = \mathcal{O}(y^{1+(N+1)\delta} M(y)^{(N+1)M})$. $\square$

Remark 9.6. Applied to $\varphi(x) = x^2(1 + \log x)$ ($\delta = 1$, $M = 1$) the theorem re-proves the remainder $\mathcal{O}(y^{N+2}M^{N+1})$ of (8.5) with a proof that uses no analyticity at all; applied to $\varphi(x) = x^2 \sin(\log x)$ ($\delta = 1$, $M = 0$) it justifies $g(y) = y - y^2 \sin L + y^3 \sin L(2 \sin L + \cos L) + \mathcal{O}(y^4)$, $L = \log y$, a perturbation whose coefficients oscillate and which no power-log expansion describes. The package's `PerturbativeInverse` produces the terms; the theorem is the user's tool for asserting the remainder.

10 Finite endpoints, infinity, and powers of the inverse

Nothing in Sections 4–8 used the sign of p or the position of the endpoint: the substitution $t = u^p$, the marker theorem at a positive point, the Euler identity and the parametric implicit-function argument only need $p \neq 0$ and $u \rightarrow 0^+$. This section makes the bookkeeping explicit.

10.1 Local variables

Let f be given near x_0 , and let $u \rightarrow 0^+$ be the local variable of x :

$$u = \begin{cases} x - x_0 & x \rightarrow x_0^+ \text{ (Direction } \rightarrow \text{ "FromAbove")}, \\ x_0 - x & x \rightarrow x_0^-, \\ 1/x & x \rightarrow +\infty, \\ -1/x & x \rightarrow -\infty. \end{cases} \quad (10.1)$$

Substituting into f and expanding (Section 3) gives a germ $y_0 + au^p(1 + \sum u^{\delta_i} B_i(\log u))$ in the sense of (4.1), where $y_0 = \lim f$ is finite when $p > 0$; when the expansion has negative powers the limit is $y_0 = \pm\infty$ (sign of a), $p < 0$, and any constant term is a correction with gap $-p$. The local variable of y is $w = |v|$, $v = y - y_0$ or $v = y$:

$$w = \begin{cases} y - y_0 & a > 0, p > 0, \\ -(y - y_0) & a < 0, p > 0, \\ 1/|y| & p < 0. \end{cases} \quad (10.2)$$

The uniformizer $z = (v/a)^{1/p}$ tends to 0^+ in all cases, and $w = |a|z^{|p|}$.

10.2 The theorem at an arbitrary endpoint

Theorem 10.1 (Signed leading power). *Theorems 4.1, 6.1, 7.1 and 8.2 hold verbatim for the germ (4.1) with $p < 0$: the unique branch with $u \rightarrow 0^+$ is $u = g(y)$ with $g \sim z$, g^r has the expansion (6.1) convergent for small z (i.e. for $|y|$ large), and the remainder after the exponent cutoff q in $w = 1/|y|$ is $\mathcal{O}(w^{(r+\sigma^*)/|p|} M(w)^{k^*})$.*

Proof. The derivative $f'(u) = apu^{p-1}(1 + o(1))$ has a constant sign, so f is monotone on $(0, \varepsilon)$ with $f \rightarrow \pm\infty$; the inverse exists on a neighbourhood of $\pm\infty$ and $u \sim z$. The proof of Theorem 6.1 is a computation at a fixed positive $q = v/a$ and never uses the sign of p . The parametric equation (7.2) has $\partial_w \mathcal{E}(1, 0) = p \neq 0$ regardless of sign, and the specialisation $T_{ij} = z^{\delta_i} L^j \rightarrow 0$ as $z \rightarrow 0^+$ is the same. The remainder theorem is Lemma 3.2. $\square$

In the variable y the blocks of g^r read $(v/a)^{(r+\sigma)/p} C_\sigma^{(r)}(\frac{1}{p} \log(v/a))$, with negative exponents of y when $p < 0$; the exponent of w is $(r + \sigma)/|p|$, which is the quantity the cutoff refers to.

10.3 Powers of the inverse and the original variable

The observable power r in (6.1) gives directly the expansion of u^r . In terms of x :

$$\begin{aligned} (x - x_0)^r &= g(y)^r (x \rightarrow x_0^+), & (x_0 - x)^r &= g(y)^r (x \rightarrow x_0^-), \\ x^r &= g(y)^{-r} (x \rightarrow +\infty), & x^r &= (-1)^r g(y)^{-r} (x \rightarrow -\infty), \end{aligned} \quad (10.3)$$

so the expansion of x^r at infinity is (6.1) with r replaced by $-r$ ("Power" $\rightarrow$ **r** in the package). For $r = 1$ and a finite endpoint, $x = x_0 \pm g(y)$. Non-integer r is allowed only when the quantity being raised is positive, i.e. for $x \rightarrow x_0^+$ and $x \rightarrow +\infty$.

10.4 Examples

Example 10.2 (The Wright omega function). $f(x) = x + \log x$ as $x \rightarrow +\infty$: $u = 1/x$, $f = u^{-1} - \log u = u^{-1}(1 - u \log u)$, so $a = 1$, $p = -1$, one gap $\delta = 1$ with $B = -L$, and $y_0 = +\infty$. The uniformizer is $z = 1/y$, $L = -\log y$, and (6.1) with $r = -1$ gives $x = g^{-1}$:

$$x = y - \log y + \frac{\log y}{y} + \frac{\frac{1}{2} \log^2 y - \log y}{y^2} + \frac{\frac{1}{3} \log^3 y - \frac{3}{2} \log^2 y + \log y}{y^3} + \mathcal{O}\left(\frac{\log^4 y}{y^4}\right), \quad (10.4)$$

the classical expansion of the Wright omega function $\omega(y) = W(e^y)$ [10, 11]; the blocks are the Lambert polynomials $P_n(\log y)$ of [11]. At $y = 10^6$ the four displayed terms differ from `ProductLog[Exp[10^6]]` by $4.8 \cdot 10^{-21}$.

Example 10.3 (Reciprocal corrections). $f(x) = x + 1/x$ at $+\infty$: $u = 1/x$, $f = u^{-1}(1 + u^2)$, $p = -1$, gap 2, constant coefficient. The inverse is the larger root of $x^2 - yx + 1 = 0$, $x = (y + \sqrt{y^2 - 4})/2 = y - y^{-1} - y^{-3} - 2y^{-5} - \dots$ (the coefficients are Catalan numbers), which (6.8) reproduces.

Example 10.4 (Half-integer gaps). $f(x) = x + \sqrt{x}$ at $+\infty$: $f = u^{-1}(1 + u^{1/2})$, gap $\frac{1}{2}$. The inverse is $x = (\sqrt{4y + 1} - 1)^2/4 = y - \sqrt{y} + \frac{1}{2} - \frac{1}{8\sqrt{y}} + \frac{1}{128y^{3/2}} + \mathcal{O}(y^{-2})$; the constant term $\frac{1}{2}$ is the block of weight $2 \cdot \frac{1}{2}$, i.e. exponent 0 in $w = 1/y$, and $y_0 = +\infty$ does not prevent it.

Example 10.5 (A finite endpoint from below). $f(x) = x - x^2$ near $x_0 = 0$ with $x \rightarrow 0^-$: $u = -x$, $f = -u - u^2$, $a = -1$, $p = 1$, $y_0 = 0$, and $w = -y$ (the image is $y < 0$). The inverse is $u = -y - y^2 - 2y^3 - \dots$, i.e. $x = y + y^2 + 2y^3 + \mathcal{O}(y^4)$, the branch of $x = (1 - \sqrt{1 - 4y})/2$ continued to negative y .

Example 10.6 (A pole). $f(x) = 1 + 1/x$ near $x_0 = 0^+$: $f = u^{-1}(1 + u)$, $p = -1$, $y_0 = +\infty$, and $x = 1/(y - 1) = y^{-1} + y^{-2} + \dots$ with remainder $\mathcal{O}(y^{-3})$ after two terms.

Example 10.7 (The second question, inverted). For $F(x) = (1 + x + x^{\sqrt{2}})^{\sqrt{2}}$ at $+\infty$ the forward germ is $u^{-2}(1 + \sqrt{2}u^{\sqrt{2}-1} + \sqrt{2}u^{\sqrt{2}} + \dots)$ from (1.5), with gaps in the semigroup generated by $\sqrt{2} - 1$ and $\sqrt{2}$, $p = -2$, $y_0 = +\infty$, $z = y^{-1/2}$. The inverse is

$$\begin{aligned} x &= \sqrt{y} - \frac{1}{\sqrt{2}} y^{(2-\sqrt{2})/2} - \frac{1}{\sqrt{2}} y^{(1-\sqrt{2})/2} \\ &\quad + \left(\frac{3}{4} - \frac{1}{2\sqrt{2}}\right) y^{(3-2\sqrt{2})/2} + \left(1 - \frac{5}{3\sqrt{2}}\right) y^{(4-3\sqrt{2})/2} + \mathcal{O}(y^{(5-4\sqrt{2})/2}), \end{aligned} \quad (10.5)$$

the terms being ordered by decreasing exponent $(r + \sigma)/2$ with $r = -1$. Because the forward germ is itself an infinite expansion, the forward remainder is transported by Theorem 9.1 and caps the attainable cutoff; the package raises the forward order automatically until the cap exceeds the request. Numerically, with seven blocks at $y = 10^{16}$ the error is $4 \cdot 10^{-11}$ against a leading term 10^8 (Table 1).

11 Worked examples

All expansions in this section were produced by the package and checked by the residual composition of Section 9 and numerically; the displayed logarithms are log of the indicated argument.

11.1 The two examples of the first question, once more

For $f_1 = x + x^2(1 + \log x)$ the package call `AsymptoticInverse[x + x^2 (1 + Log[x]), {x, 0}, {y, 6}]` returns (1.3) with the remainder `PowerLogRemainder[y, 6, 5]`; the sixth block is

$$\begin{aligned} P_5(L) &= -42L^5 - \frac{3727}{12}L^4 - \frac{5365}{6}L^3 - 1256L^2 - \frac{5177}{6}L - \frac{2789}{12} \\ &= -\frac{1}{12}(1+L)(504L^4 + 3223L^3 + 7507L^2 + 7565L + 2789), \end{aligned}$$

and the seventh $P_6 = \frac{1}{60}(1+L)(7920L^5 + 63852L^4 + 200833L^3 + 308537L^2 + 231883L + 68307)$. With rational exponents the same object is available as `SeriesData[y, 0, {1, -1 - Log[y], 3 + 5 Log[y] + 2 Log[y]^2, ...}, 1, 6, 1]`, on which the built-in series arithmetic works (with the caveat that its $O[y]^6$ hides the factor $M(y)^5$).

For $f_2 = x + x^{\sqrt{2}}$, `SeriesTermGoal -> 4` returns exactly the four terms requested in the question with remainder `PowerLogRemainder[y, 4 Sqrt[2] - 3, 0]`; the exponent cutoff that produces the same result is $4\sqrt{2} - 3$.

11.2 Nonunit leading power

$f(x) = 3x^2(1 + x(1 + \log x))$: $a = 3$, $p = 2$, $\delta = 1$, $B = 1 + L$. With $z = \sqrt{y/3}$ and $L = \log z = \frac{1}{2}\log(y/3)$, (6.3) gives

$$g(y) = z - \frac{z^2}{2}(1+L) + \frac{z^3}{8}(5L^2 + 12L + 7) - \frac{z^4}{48}(\mathcal{D} + 6)(\mathcal{D} + 8)(1+L)^3 + \mathcal{O}(z^5 M^4), \quad (11.1)$$

where $\frac{1}{48}(\mathcal{D} + 6)(\mathcal{D} + 8)(1+L)^3 = (1+L)^3 + \frac{7}{8}(1+L)^2 + \frac{1}{8}(1+L)$. Two ingredients are easy to forget: the factor $p^{-n} = 2^{-n}$ and the logarithm $L = \frac{1}{2}\log(y/3)$, not $\log y$. In the variable y the expansion reads

$$\begin{aligned} g(y) &= \frac{\sqrt{y}}{\sqrt{3}} - \frac{y}{3}\left(\frac{1}{2} + \frac{\log(y/3)}{4}\right) + \frac{y^{3/2}}{3\sqrt{3}}\left(\frac{7}{8} + \frac{3}{4}\log\frac{y}{3} + \frac{5}{32}\log^2\frac{y}{3}\right) \\ &\quad - \frac{y^2}{9}\left(2 + \frac{39}{16}\log\frac{y}{3} + \frac{31}{32}\log^2\frac{y}{3} + \frac{1}{8}\log^3\frac{y}{3}\right) + \mathcal{O}(y^{5/2} M^4). \end{aligned}$$

11.3 Several gaps

Irrational and logarithmic gaps together. $f(x) = x + x^{\sqrt{2}} + x^2(1 + \log x)$ has gaps $\delta_1 = \sqrt{2} - 1$ and $\delta_2 = 1$. Below the exponent 3 the support is $\{1, \sqrt{2}, 2\sqrt{2} - 1, 2, 3\sqrt{2} - 2, 1 + \sqrt{2}, 4\sqrt{2} - 3, 2\sqrt{2}\}$ and

$$\begin{aligned} g(y) &= y - y^{\sqrt{2}} + \sqrt{2}y^{2\sqrt{2}-1} - y^2(1+L) - \frac{6-\sqrt{2}}{2}y^{3\sqrt{2}-2} + y^{1+\sqrt{2}}((\sqrt{2}+2)L + \sqrt{2}+3) \\ &\quad + \left(\frac{17\sqrt{2}}{3} - 4\right)y^{4\sqrt{2}-3} - y^{2\sqrt{2}}\left((3\sqrt{2}+5)L + 5\sqrt{2} + \frac{13}{2}\right) + \mathcal{O}(y^3 M^2), \quad L = \log y, \end{aligned}$$

the mixed blocks coming from (6.4): the index $(1, 1)$ gives $(\mathcal{D}+2+\sqrt{2})(1+L) = (\sqrt{2}+2)(1+L)+1$. The remainder is the block $y^3(2L^2 + 5L + 3)$ of the index $(0, 2)$, of degree 2.

Two irrational gaps. $f(x) = x + x^{\sqrt{2}}(1 + \log x) + 2x^{\sqrt{3}}$ has gaps $\sqrt{2} - 1$ and $\sqrt{3} - 1$, whose sums are the numbers $k_1(\sqrt{2} - 1) + k_2(\sqrt{3} - 1)$; the package represents them by `RootReduce` (for instance $\sqrt{2} + \sqrt{3} = \text{Root}[1 - 10 \#^2 + \#^4 \&, 4]$) and displays them with `ToRadicals`. Below the exponent 3 there are eleven blocks; the residual check passes only when the coefficients, which involve $\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}$, are canonicalised with `RootReduce`: this is precisely the zero-recognition defect that made one test of report 01 fail (Section 14).

11.4 Automatically expanded forward functions

$\sin x + x^2 \log x$ is not a finite power-log sum. The evaluator of Section 3.2 expands it to $x + x^2 \log x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \mathcal{O}(x^7)$, records $(\rho, k) = (7, 0)$, and the inverse below the exponent 4 is

$$g(y) = y - y^2 \log y + y^3 \left(\frac{1}{6} + \log y + 2 \log^2 y \right) + \mathcal{O}(y^4 M^3);$$

here the y^3 block combines $(\mathcal{D} + 4)L^2/2 = 2L^2 + L$ from the index $(2, 0)$ (gap 1, $B_1 = L$) with $+\frac{1}{6}$ from the index $(0, 1)$ (gap 2, $B_2 = -\frac{1}{6}$). Similarly $\tan x$ gives `arctan: SeriesTermGoal -> 4` yields $y - \frac{1}{3}y^3 + \frac{1}{5}y^5 - \frac{1}{7}y^7$, the forward order being raised automatically until the transported remainder allows the fourth term.

11.5 Symbolic data

With symbolic coefficients the exponents remain numeric and everything goes through under assumptions that make the leading coefficient's sign and the reality of the coefficients provable: $f = ax + bx^2$ with $a > 0$, $b \in \mathbb{R}$ gives $g = y/a - by^2/a^3 + 2b^2y^3/a^5 + \mathcal{O}(y^4)$. With a symbolic exponent the ordering of the support is undecidable in general and depth truncation (Theorem 8.6) is used: $f = x + x^\alpha$ with $\alpha > 1$ gives, at depth 3,

$$g = y - y^\alpha + \alpha y^{2\alpha-1} - \frac{\alpha(3\alpha-1)}{2} y^{3\alpha-2} + \mathcal{O}(y^{4\alpha-3}),$$

which is (6.9) with a symbolic α .

11.6 Powers and logarithms of the inverse

For $f = x + x^2$, (6.1) with $r = 2$ gives $g^2 = y^2 - 2y^3 + 5y^4 - 14y^5 + \dots$ (Catalan numbers again), with $r = -1$ it gives $1/g = y^{-1} + 1 - y + 2y^2 - 5y^3 + \dots$, and (6.2) gives $\log g = \log y - y + \frac{3}{2}y^2 - \frac{10}{3}y^3 + \dots$. These expansions are exact consequences of the same coefficient formula and cost nothing extra; expanding `Normal[g]^2` instead would silently lose the remainder information.

12 Boundaries of the theory

The finite power-log class is closed under the operations of Theorem 3.3 and under inversion of germs with a monomial leading block. Each of the following situations leaves it, and each requires a specific extension, described here to the extent needed to recognise it and to say what the extension must provide.

12.1 A leading logarithm: Lambert W and the log log scale

If the leading block is $ax^p(-\log x)^b$ with $b \neq 0$, the equation $ax^p(-\log x)^b = y$ is solved exactly by a Lambert W function. With $t = -\log x \rightarrow +\infty$ and $Y = y/a$ it reads $t^b e^{-pt} = Y$, i.e.

$(-\frac{p}{b}) t e^{-pt/b} = -\frac{p}{b} Y^{1/b}$, so

$$x = \exp\left[\frac{b}{p} W_k\left(-\frac{p}{b} Y^{1/b}\right)\right], \quad k = -1 \ (b > 0), \quad k = 0 \ (b < 0), \quad (12.1)$$

the branch being forced by $t \rightarrow +\infty$: for $b > 0$ the argument is small and negative and only $W_{-1} \rightarrow -\infty$ gives $t \rightarrow +\infty$ (the branch W_0 describes the other real preimage, near $x = e^{-b/p}$, where $ax^p(-\log x)^b$ has its maximum); for $b < 0$ the argument is large and positive and $W_0 \rightarrow +\infty$. For instance the inverse of $x \log(1/x)$ is $x = -y/W_{-1}(-y)$. Expanding W_{-1} for small argument [10] gives, with $\Lambda = \log(a/y) \rightarrow \infty$ and $\Lambda_2 = \log \Lambda$,

$$x = \left(\frac{y}{a}\right)^{1/p} \left(\frac{\Lambda}{p}\right)^{-b/p} \left[1 - \frac{b^2}{p\Lambda} \log \frac{\Lambda}{p} + \mathcal{O}\left(\frac{\Lambda_2^2}{\Lambda^2}\right)\right], \quad (12.2)$$

a series in the two generators $1/\Lambda$ and Λ_2 multiplying the leading monomial. Perturbations $x^\delta B(\log x)$ of such a core can be handled by the general-core formula (5.5) with ϕ the function (12.1), and the result is a convergent expansion in powers of $\phi(y)$ with coefficients that are rational functions of $t = -\log \phi(y)$, because $\phi'(y) = [apx^{p-1}t^{b-1}(pt - b)]^{-1}$ at $x = \phi(y)$ introduces the denominator $pt - b$. Re-expanding those coefficients in $1/\Lambda$ and Λ_2 produces the “nested logarithm” transseries of [11, 8]. An implementation needs a coefficient algebra of rational functions of t (or of polynomials in Λ_2 and $1/\Lambda$), a second truncation policy for the logarithmic depth, and the corresponding closure and remainder theorems; the package reports such inputs as `Failure["LogarithmicLeadingTerm"]` rather than pretending to be a transseries engine.

12.2 Zero gaps: corrections of relative size $1/\log$

For $f(x) = x + x/\log x$ the correction has power gap 0 and relative size $1/\log x$. All corrections of the inverse then sit at the same power y with increasing negative powers of $\log y$, the index set I_H is infinite, and the exponent cutoff has no meaning. The right coordinate is $\tau = 1/\log y$: with $g = yW$ the equation becomes $W(1 + \tau/(1 + \tau \log W)) = 1$, analytic near $(W, \tau) = (1, 0)$, so W is an ordinary convergent power series in τ , $W = 1 - \tau + \tau^2 - 2\tau^3 + \frac{7}{2}\tau^4 - \dots$, i.e. $g(y) = y(1 - 1/\log y + 1/\log^2 y - 2/\log^3 y + \dots)$. The general-core formula produces these terms, but the scale is the reciprocal-logarithm scale, not the power-log scale, and a package extension needs τ as a second generator with its own valuation.

12.3 Flat terms

$e^{-1/x}$ is smaller than every power of x ; a germ $x + e^{-1/x}$ has the same power-log expansion as x , and its inverse differs from y by $-e^{-1/y}(1 + o(1))$ (by Theorem 9.1 the difference is $\mathcal{O}(e^{-1/y})$, and the marker formula gives $y - e^{-1/y} + y^{-2}e^{-2/y} + (y^{-3} - \frac{3}{2}y^{-4})e^{-3/y} + \dots$). Such terms live in exponential sectors of a transseries; the power-log expansion is blind to them, and the package rejects exponentials of unbounded arguments instead of dropping them.

12.4 Oscillatory coefficients

$x^2 \sin(\log x)$ is $\mathcal{O}(x^2)$ with all derivatives of the right size, so Theorem 9.5 applies and the marker formula gives a valid expansion $y - y^2 \sin L + y^3 \sin L(2 \sin L + \cos L) + \mathcal{O}(y^4)$ whose “coefficients” are bounded oscillating functions of $L = \log y$. They are not polynomials in L , the coefficient algebra is different, and the sharp remainder theory of Section 8 (which uses degrees) does not apply; only the tame theorem does. The package’s `PerturbativeInverse` generates the terms and leaves the remainder to the user.

12.5 Symbolic exponents and nonuniformity

When a gap δ is a symbol, the order of the weights $k \cdot \delta$ is a question about the parameter: for two gaps δ_1, δ_2 the relative order of $2\delta_1$ and δ_2 depends on whether $\delta_2 < 2\delta_1$. Exponent truncation therefore requires a proof of every comparison from the assumptions, which the package attempts (`FullSimplify` under `Assumptions`) and refuses when it fails; depth truncation (Theorem 8.6) needs only the positivity of the gaps. Nothing is uniform in a gap tending to 0: the convergence radius (6.10) of the $x + x^\alpha$ inverse tends to 0 as $\alpha \downarrow 1$, and at $\alpha = 1$ the inverse is $y/2$, not $y + o(y)$.

12.6 Nested powers of the logarithm

Non-integer powers $(\log x)^s$ or $\log \log x$ in a correction, and negative powers of $\log x$ in a coefficient, all leave the polynomial coefficient algebra; a correction $x^\delta (\log x)^{1/2}$ is admissible in the sense of Theorem 9.5 (with $M = 1$) but its inverse expansion has coefficients in $\mathbb{R}[L^{1/2}]$. Each such extension is a change of the coefficient ring with the same exponent bookkeeping; the Euler identity and the composition recurrence survive (they only use ∂_L), and what has to be added is the closure of the new coefficient ring under the operations and the corresponding remainder classes.

13 The package

The package `AsymptoticInverse` (directory `AsymptoticInverse/` of the repository; usage in its `README.md`) implements Sections 2–10. Its interface is

```
AsymptoticInverse[f, {x, x0}, {y, cutoff}]      (* or ..., y, SeriesTermGoal -> n
*)
AsymptoticExpansion[f, {x, x0, cutoff}]        (* or {x, x0}, SeriesTermGoal -> n
*)
```

with x_0 a real number, `Infinity` or `-Infinity`, options `Assumptions`, `Direction`, `Method` ("Lagrange" or "Newton"), "Power" (r), "Truncation" ("Exponent" or "Depth"), "InputRemainder" and "MaxTerms". Both return a `PowerLogSeries` object: `Normal` gives the finite expression, `s["Remainder"]` the descriptor `PowerLogRemainder[w,  $\beta$ , k]`, `s["FrontierTerm"]` the first omitted block, `s["Terms"]` the pairs (exponent, coefficient), `s["SeriesData"]` a `SeriesData` object when the exponents are rational, and `s["Properties"]` the full list. `InverseResidual`, `InverseNumericalCheck`, `PerturbativeInverse`, `InverseExpansionCoefficient` and `PowerLogModel` are the auxiliary functions.

13.1 Data structures and exact comparisons

A jet is a sorted list of pairs $\{\sigma, C_\sigma\}$, the exponent σ an exact real number and C_σ a polynomial in a private symbol ℓ standing for L . Exponents are canonicalised with `RootReduce` when they are algebraic (so that $\sqrt{2} + \sqrt{3}$ and $\sqrt{5 + 2\sqrt{6}}$ become the same `Root` object) and with `FullSimplify` otherwise (exact transcendental exponents such as π are admitted when every comparison they enter can be decided). Two exponents are equal if and only if their canonical forms are identical; their order is decided by `Less` on exact numbers (which is exact for distinct algebraic numbers) with `FullSimplify` of the difference as a fallback, and an undecidable comparison is a `Failure`, never a numerical guess. Inexact (`Real`) input is rejected: a floating-point 1.41421 is a rational number, and inverting $x + x^{1.41421}$ is a different problem (Section 12).

Coefficients are canonicalised too: algebraic numbers with `RootReduce`, logarithms of positive rationals into sums of logarithms of primes, symbolic parameters with `Simplify` under the assumptions. Without this, exactly vanishing residual coefficients such as $-2\sqrt{2} - 2\sqrt{3} + 2\sqrt{5 + 2\sqrt{6}}$ or $-\frac{1}{16} \log^2 4 - \frac{1}{64} \log 2 \log 32 + \frac{1}{64} \log 8 \log 128$ are not recognised as zero by `Simplify`, which is what caused the two test failures among the nine reports (Section 14).

13.2 The engines

Forward. The evaluator of Section 3.2 is a structural recursion returning triples (T, P, D) . Finite power-log sums are recognised as exact ($P = \infty$) without any series; otherwise the working order is raised until the requested precision is reached, and the first omitted block is computed when it lies within reach.

Inverse, Lagrange. The forward germ is normalised (Section 4); the multi-indices of weight $< H$ are enumerated by a depth-first search with exact comparisons ((8.1); the one-step boundary is formed at the same time); for each index the polynomial loop

```
q = Expand[Times @@ MapThread[Power, {polys, k}]];
Do[q = Expand[D[q, ell] + (r + w + p j) q], {j, 1, n - 1}];
q = (-1)^n r q / (p^n Times @@ (Factorial /@ k))
```

evaluates (6.1); equal weights are merged and the blocks of weight σ_* on the boundary are summed into the frontier term. For `SeriesTermGoal -> n` the inclusive weight bound is raised through successive boundary weights until n nonzero blocks exist.

Inverse, Newton. The normalised equation (6.11) and its derivative are evaluated with the composition recurrence (Lemma 2.7), the derivative's reciprocal by the geometric series of a unit with constant leading term, and the Newton step is repeated until the residual vanishes below H ; the number of iterations is logarithmic in H/δ_* . The two engines are compared in the test suite.

Assembly. Blocks $z^{r+\sigma} C(L)$ are converted to $(v/a)^{(r+\sigma)/p} C(\log(v/a)/p)$, the sign and shift of (10.3) are applied, and the remainder class is the coarser of the truncation remainder and the transported forward remainder.

13.3 Checks

`InverseResidual` composes the stored forward model with the stored blocks in the jet algebra and reports whether the normalised residual $f(G)/(az^p) - 1$ vanishes below the residual cutoff; with a larger cutoff it exhibits the first omitted block with the factor $-p$ of (9.5). This check is independent of the coefficient formula (it uses only the composition recurrence). `InverseNumericalCheck` solves $f(x) = y_1$ with `FindRoot` at high precision from the expansion value and reports the error, the remainder scale and their ratio (Table 1); it is evidence, not a certificate. The 49 tests in `Tests/AsymptoticInverse.wlt` cover the examples of this article, resonances, several gaps, symbolic data, the two methods, all endpoint types, automatically expanded inputs, the checks, and the rejected inputs; all pass on Wolfram 15.0.1.

13.4 Cost

The number of retained multi-indices is at most $\prod_i (1 + \lfloor H/\delta_i \rfloor)$, and each coefficient costs $|\mathbf{k}| - 1$ applications of a bidiagonal operator to a polynomial of degree $D(\mathbf{k})$. The dominant cost in practice is the canonicalisation of algebraic numbers: expansions with two incommensurable gaps below the exponent 3 take a few seconds, whereas rational exponents take milliseconds. The Newton engine performs $\mathcal{O}(\log(H/\delta_*))$ compositions whose cost is governed by the sparse products; it is competitive for a single gap and slower for many gaps.

14 The nine reports

Nine independently written reports on the first question [12] preceded this article; each contains an article of 22–30 pages, a Wolfram Language package, native tests, and a Python verification record, and none of them had been run in a Wolfram kernel by its authors. Their detailed comparison is in `reports/COMPARISON.md`; this section summarises what they agree on, what each adds, and what was wrong.

14.1 Agreement

All nine treat the finite germ (4.1) (report 07 with $p = 1$ only), select the real branch by $x/(y/a)^{1/p} \rightarrow 1$, prove the coefficient formula (6.1) with exactly the same normalization through the marker form of Lagrange–Bürmann inversion, prove convergence (four by the parametric implicit function theorem, five by Rouché’s theorem with Cauchy estimates), prove a remainder theorem with the logarithmic factor, derive the transport of forward remainders, and compute the same expansions (1.3)–(1.4) with the Catalan and Fuss–Catalan laws. All represent expansions as sparse exact jets, reject floating-point input, and provide a residual check.

14.2 Individual contributions absorbed here

- Report 01: the residue proof of the marker theorem and its formal clause; the g^r and $\log g$ formulas; the `InputRemainder` interface with cutoff cap; the $x + x/\log x$ analysis.
- Report 02: the observable power r built into the main formula and implemented; the Picard fixed-point engine; the explicit marker-disc convergence criterion; the partial resummation in $y \log y$.
- Report 03: the exact degree and leading-coefficient formula; the contraction proof of convergence; the search bound $\lceil B/\delta_* \rceil \delta_*$ for the frontier.
- Report 04: the first omitted block as an asymptotic equality with its complete polynomial; the successive-power engine; the condition number.
- Report 05: the tame near-identity theorem (here Theorem 9.5, with a complete proof); Picard iteration; global bounds for both examples.
- Report 06: the composition recurrence (Lemma 2.7); the Newton engine in the jet algebra; the rising-factorial closed form (6.8).
- Report 07: the explicit conditional majorant $\mathcal{Q}_r$ (Theorem 7.3) and its cutoff form; the negative-side germ by reflection.
- Report 08: the explanation of the complex germ (1.2); the Lambert- W core (12.1) and the $\log \log$ scale; the finite-point slope bound; flat perturbations.

- Report 09: depth truncation with symbolic exponents; the harmonic-number formula (6.7); the explicit tail majorant for the log example.

14.3 Errors found

Report 07 asserts that a nonunit leading power forces an infinite binomial expansion; the substitution $t = x^p$ in Theorem 6.1 shows that the coefficient formula stays finite and exact. Reports 07 and 08 state remainder degrees $(d[\sigma_*/\delta], Nd)$ that are valid but can be far from the exact degree of the frontier block. Report 05's proof of the tame theorem omits the inductive estimates of Step 4 of Theorem 9.5. Two residual checkers (reports 01 and 04) rely on `Simplify` to recognise zero and fail on denested radicals and on logarithms of composite integers.

14.4 Native execution

Running the nine test suites on Wolfram 15.0.1 gave: 01: 35/36; 02: 95/95; 03: 34/34; 04: 34/35; 05: 38/38; 06: 33/33; 07: 34/34; 08: 40/40; 09: 35/35. The two failures are the zero-recognition defects just mentioned, in residual checkers, not wrong expansions. Every package therefore computes what its article claims, which is a strong independent confirmation of the formula (6.1) and of the two answers. What none of them does, and what motivated the present package, is to treat finite endpoints from both sides and infinity through signed leading powers, to expand arbitrary forward functions with a rigorous remainder (which is the second question), to track precision and logarithmic degree through compositions, and to return `SeriesData` when the exponents allow it.

References

- [1] V. Reshetnikov, *How to get an asymptotic of the real-valued branch of the inverse function?*, Mathematica Stack Exchange, question 236367, December 2020. <https://mathematica.stackexchange.com/questions/236367/>
- [2] V. Reshetnikov, *Asymptotic expansion for a function containing irrational exponents*, Mathematica Stack Exchange. (Local copy in `docs/`.)
- [3] NIST Digital Library of Mathematical Functions, §1.10(vii) Lagrange inversion, §4.13 Lambert W -function. <https://dlmf.nist.gov/>
- [4] I. M. Gessel, *Lagrange inversion*, J. Combin. Theory Ser. A **144** (2016), 212–249. arXiv:1609.05988.
- [5] E. Surya, L. Warnke, *Lagrange inversion formula by induction*, arXiv:2305.17576.
- [6] G. A. Edgar, *Transseries for beginners*, Real Anal. Exchange **35** (2009/10), 253–309. arXiv:0801.4877.
- [7] G. A. Edgar, *Transseries: composition, recursion, and convergence*, arXiv:0909.1259.
- [8] J. van der Hoeven, *Transseries and Real Differential Algebra*, Lecture Notes in Mathematics 1888, Springer, 2006.
- [9] B. Salvy, J. Shackell, *Symbolic asymptotics: multiseries of inverse functions*, J. Symbolic Comput. **27** (1999), 543–563.
- [10] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, D. E. Knuth, *On the Lambert W function*, Adv. Comput. Math. **5** (1996), 329–359.

- [11] V. Reshetnikov, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and the inversion of rapidly growing functions* and *Combinatorial transseries and their inverses*, ProveIt repository, directory `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries`, September 2026. <https://github.com/VladimirReshetnikov/ProveIt>
- [12] The nine reports in the directory `reports/` of this repository (`reports/COMPARISON.md`), 7 September 2026.
- [13] Wolfram Research, Wolfram Language documentation: `Series`, `SeriesData`, `InverseSeries`, `InverseFunction`, `Asymptotic`, `AsymptoticSolve`, `RootReduce`, `ProductLog`. <https://reference.wolfram.com/language/>